

# **EEA5.0 ECU bootloader and programming specification**

## **EEA5.0 ECU bootloader 及刷新规范**

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# 1 Introduction/介绍

## 1.1 Scope/范围

This document specifies detailed methods and requirements on Unified Diagnostic Service (UDS)-based software reprogramming of Chery passenger vehicle ECUs. The software reprogramming of Chery passenger vehicle ECUs shall comply with this document. This document is applicable to reprogramming of in-vehicle ECUs which are connected with an external diagnostic device. For programming between the Micro Processor Unit (MPU) and Micro Controller Unit (MCU) in an ECU and of the proprietary components in an ECU (not exposed to external diagnostic devices), this document is not mandatory

本规范规定了奇瑞乘用车 ECU 基于 UDS 的软件通用重编程的详细方法和要求。奇瑞乘用车的 ECU 软件重编程功能开发应符合本规范。本规范适用于外部诊断设备和车内 ECU 之间的重编程规范，部件内部的 MPU 和 MCU 之间刷写以及部件的私有部件（不对外部诊断设备呈现的部件）不做强制要求。

## 1.2 References/参考文献

The following reference documents are indispensable for the application of this document.

以下引用文件对于本文件的应用是必不可少的

《Q/SQR E8-20-2022 Vehicle Unified diagnostic services (UDS) Standard》

《Vehicle UDS Diagnostic Standard on CAN implementation (UDSonCAN) 》

《Vehicle UDS Diagnostic Standard on IP implementation (UDSonIP) 》

《EEA5.0 ECU 诊断调查表》

《EEA5.0 ECU 27 服务算法规范》

《EEA5.0 ECU 软件刷写签名算法规范》

## 1.3 Terms and Definitions/术语和定义

Terms defined in ISO 14229-1 and ISO 13400-2 and the following terms and definitions are applicable to this document.

The diagnostic message is expressed in a character string starting with the dollar sign (\$). For example, \$10 01 indicates a diagnostic message of the diagnostic service whose SID is 0x10 and the diagnostic sequence is 10h 01h (h indicates a hexadecimal number).

ISO 14229-1, ISO 13400-2 确定的以及下列术语和定义适用本规范。

对于诊断消息描述，采用“\$”开头的字符串表示，例如\$10 01，表示 UDS 的诊断服务 SID=0x10 的诊断消息，诊断序列为 10h 01h（h 代表十六进制）。

| Term/术语                    | Abbreviation/缩略语 | Definition/解释 |
|----------------------------|------------------|---------------|
| Unified Diagnostic Service | UDS              | 统一诊断服务        |
| Electronic Control Unit    | ECU              | 电子控制单元        |
| Service Identifier         | SID              | 服务标识符         |
| Negative Response Code     | NRC              | 否定响应码         |
| Diagnostic Trouble Code    | DTC              | 诊断故障码         |

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| Term/术语                            | Abbreviation/缩略语 | Definition/解释 |
|------------------------------------|------------------|---------------|
| Data Identifier                    | DID              | 数据标识符         |
| Routine Identifier                 | RID              | 例程标识符         |
| Diagnostics Over Internet Protocol | DoIP             | 基于以太网网络协议的诊断  |
| Diagnostics Over CAN               | DoCAN            | 基于 CAN 协议的诊断  |
| Tester                             | Tester           | 测试仪，即外部诊断设备   |
| Random Access Memory               | RAM              | 随机存取存储器       |
| Most significant bit               | MSB              | 最高有效位         |
| Least significant bit              | LSB              | 最低有效位         |
| Micro Controller Unit              | MCU              | 微控制单元         |
| Micro Processor Unit               | MPU              | 微处理单元         |

### 1.3.1 Logical Block and Segment/逻辑块和软件段

A logical block is formed by one or more segments. The flash space occupied by logic blocks and segments cannot overlap. During software upgrade of a logical block containing discontinuous software segments, if the Tester erases the entire logical block according to the start address and length of the logical block, the ECU needs to skip, fill, or protect related parts. If the Tester performs multiple erasures based on the start address and length of a software segment, the ECU shall support multiple erasures at a time. The ECU needs to support the preceding two erasing methods.

逻辑块（Logical block）由一个或者多个软件段（Segment）组成。软件逻辑块和软件段占用的 Flash 空间之间不能有交叉、重叠。在进行软件升级一个包含不连续软件段的逻辑块时，如果 Tester 按照逻辑块的起始地址和长度进行整个逻辑块的擦除，擦除的时候 ECU 需要自己跳过、填充或者保护处理；如果 Tester 按照软件段的起始地址和长度进行多次擦除，ECU 需要支持一次性执行多次擦除。ECU 需要支持两种擦除方式。

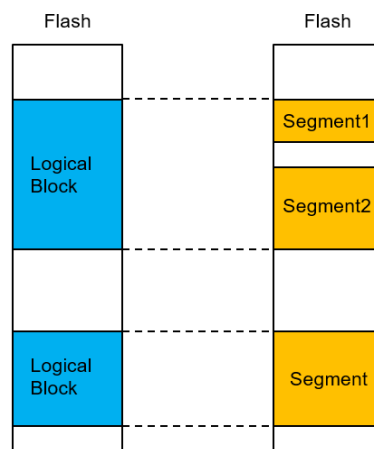


Figure 1: Relationship between logic blocks and segments/逻辑块和软件段关系

|                                                 |                  |                  |
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Note: A logical block maps a programming file, and a programming file can contain multiple segments. A complete ECU programming package can contain one or more logical blocks.

说明：一个逻辑块，需要对应一个刷写文件，一个文件可以包含多个软件段。一个完整的 ECU 刷写文件包，可以包含一个或者多个逻辑块。

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## 2 Reprogramming Requirements/重编程要求

The application, calibration, and configuration data in the ECU can be reprogrammed. Corresponding data packages can be downloaded together or separately.

ECU 内部允许重编程的数据包括应用程序、标定数据和配置数据等。这些数据包可以一起下载，也可以支持独立下载。

If the ECU does not support reprogramming, a negative **NRC 12** should be returned upon receipt of the diagnostic service whose SID is 0x10 to switch to the programming session mode (sub-function not supported).

如果 ECU 不支持重编程，在接收到 SID=0x10 诊断服务切换编程会话时，应该返回否定 NRC 12（子功能不支持）。

The system can perform reprogramming in the case of reprogramming failure, interruption, timeout, reset, incomplete flash erasing, or invalid applications. The solutions include but are not limited to the following: (1) The unified StayInBoot mechanism defined in this document is supported. (2) The ECU has a secure startup mechanism. When an application is invalid (the ECU needs to have an application integrity and validity check mechanism), the ECU automatically returns to the Bootloader and supports reprogramming. (3) The ECU can automatically return to the Bootloader after a valid application fails. For example, if the watchdog resets for five consecutive times, the ECU stays in the Bootloader.

在重编程失败、被中断、发生超时、复位、Flash 擦除不完全或应用程序无效的情况下，系统能够重编程。具体方案包括但不限于如下手段：1）支持本规范定义的统一 StayInBoot 机制；2）ECU 具备安全启动机制，在 APP 无效（ECU 需要有 APP 完整性有效性检查机制）时主动回到 Bootloader，支持重编程；3）ECU 在 APP 有效但跑飞后能主动回到 Bootloader，如：看门狗连续狗咬复位 5 次，则主动停留在 Bootloader。

After receiving the message sent by the service whose SID is 0x28, all ECUs connected to the network should forbid transmitting corresponding messages. Normal communication of an ECU should be enabled after the message that enables normal communication is received from the service whose SID is 0x28 or the ECU returns to the default session mode.

所有连接到网络的 ECU 在收到 SID=0x28 的消息后，应该禁止相应的消息发送。所有 ECU 的正常通信应该在收到使能正常通信的 SID=0x28 的消息后或者 ECU 返回到默认会话之后使能。

All ECUs connecting to the network should disable the DTC setting function after receiving the "SID=0x85" message. The DTC setting function of an ECU should be enabled after the message sent by the service whose SID is 0x85 is received or the ECU returns to the default session mode.

所有连接到网络的 ECU 在收到 SID=0x85 的消息后，应该禁止 DTC 设置。所有 ECU 的 DTC 设置功能应该在收到使能 DTC 设置 SID=0x85 的消息后或者 ECU 返回到默认会话之后使能。

After the software version file is downloaded, the ECU should perform the programming dependency check. For example, whether the Bootloader version matches the application version, and whether software matches hardware. If the check fails, the ECU returns a message indicating that the check fails.

软件下载完毕后，ECU 应该进行编程依赖检查，建议进行 Bootloader 版本和 APP 版本匹配检查，软件和硬件匹配检查等，如果检查失败在进行编程依赖检查时应该返回编程依赖检查失败。

For ECUs that support DoCAN/CAN FD, the values of the **BS** and **STmin** parameters at the programming session network layer in the Bootloader are both **0**. If the ECU cannot meet the preceding time parameter requirements, the ECU reprogramming can be performed only after the diagnostic engineer's approval.

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支持 DoCAN/CANFD 的 ECU， Bootloader 中编程会话网络层参数 BS = 0 和 STmin = 0。如果 ECU 无法满足此时间参数要求， 需要取得诊断工程师的审批方可实施。

Before the reset, the ECU that supports DoIP should actively disable and clear the DoIP link within 1s. 支持 DoIP 的 ECU 复位之前， 应该在 1 秒内主动关闭和清理 DoIP 链路， 再执行复位动作。

During the interaction between the Tester and the ECU, the Tester should terminate the programming (except the processing of optional operations) when receiving a negative physical addressing response (except NRC 78) from the ECU.

Tester 和 ECU 交互过程中， 收到 ECU 响应的物理寻址否定响应（NRC 78 除外）消息时， Tester 应该终止本次编程（可选步骤的处理除外）。

During data transmission, if a \$36 message frame timeout occurs, the Tester resends the \$36 message frame. The ECU needs to correctly process the \$36 message data in this scenario according to the ISO 14229 standard without terminating the programming.

数据传输过程中， 如果某一帧\$36 消息超时错误， Tester 会重新发送此帧\$36 消息。 ECU 需要按照 ISO 14229 的标准， 正确处理这个场景的\$36 消息数据， 不要终止编程。

## 2.1 File Format/文件格式

The ECU program file should be in the CBF (Chery Binary Format) format. The ECU supplier needs to convert the S19, Hex, Bin or zip format file into the CBF format through the CBF conversion tool, which is released by Chery.

ECU 刷新文件采用 CBF（Chery Binary Format）格式， ECU 供应商需要将原始 S19、 Hex、 Bin 或者 zip 格式文件通过 CBF 转换工具转换成 CBF 格式， CBF 转换工具由奇瑞释放。

## 2.2 Diagnostic Session Management/诊断会话管理

When the ECU works in the Bootloader model, the input and output of the ECU (if any) should be kept in the security status, and the power consumption of the ECU should be reduced as much as possible. For details about the power consumption requirements of each ECU, see the specifications provided by corresponding OEMs. This section describes the conversion logic between the default, extended, and programming session modes.

ECU 工作在 BootLoader 模式时， 如果是有输入和输出的 ECU， 在 BootLoader 模式时应该保证在安全的输入输出状态， 并且尽可能降低 ECU 的功耗， 各 ECU 功耗要求见主机厂规范。 本章节描述了默认会话模式， 扩展会话模式， 编程会话模式三者之间的转换逻辑。

To ensure that the ECU can always execute reprogramming, a 20 ms delay is enabled after the Bootloader detects that an application is valid. If the ECU receives the diagnostic message of the StayInBoot function within 20 ms, the ECU enters the default session mode of the Bootloader instead of entering the application model.

为了保证 ECU 始终能够重编程， 在 BootLoader 检测到应用程序有效后启动一个 20ms 的延时， 如果 ECU 在此 20ms 内收到 StayInBoot 的诊断报文， 则不进入 APP 而是进入 BootLoader 的默认会话模式。

## 2.2.1 Application not Supporting Reprogramming/应用程序不支持编程

### 2.2.1.1 Not Supporting A and B Dual Backup/不支持 AB 区双备份

For an ECU of an embedded system (without an operating system (OS) or file system), application programming is not supported. Figure 2 shows the flowchart of the Bootloader startup and diagnostic session management of this kind of ECUs.

对于传统 ECU（不带操作系统和文件系统），通常不支持应用程序编程，其 ECU 的 BootLoader 启动流程图和诊断会话管理参考见 Figure 2 所示。

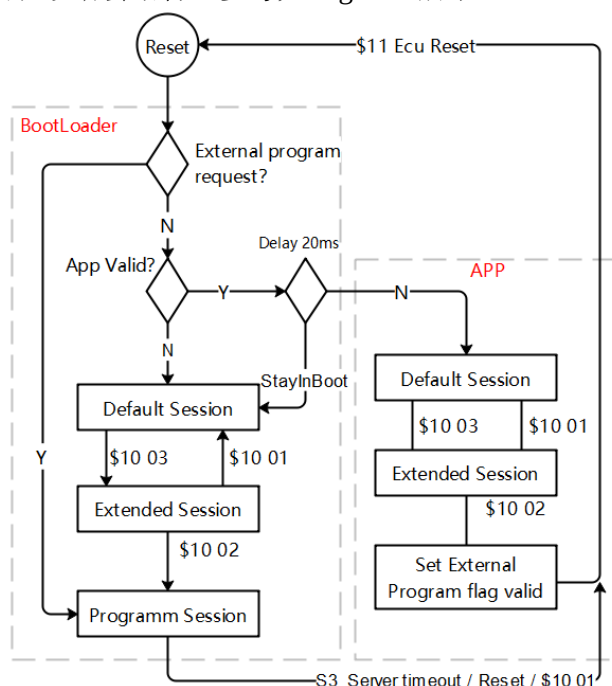


Figure 2: Process of reprogramming not supported and dual backup not supported by the application  
应用程序不支持重编程且不支持 AB 区流程

1. In the Bootloader model, after the ECU enters the programming mode, if the S3 Server timeout or reset occurs or the \$10 01 message is received, the ECU reset.

在 Bootloader 下，只要进入了编程模式，那么 S3 Server 超时，复位，以及 \$10 01，ECU 应该复位。

2. After detecting that an application is valid, the ECU executes a 20 ms delay. If the ECU receives the StayInBoot UDS diagnostic message of \$31 01 DD 01 within the delay period, the ECU directly enters the default session mode of the Bootloader. In the application model, the ECU forcibly requires the Routine Identifier (RID) that does not support 0xDD01. In the Bootloader model, the ECU forcibly requires the RID that supports 0xDD01. After the Bootloader state is successfully entered, the ECU responds with a positive response message of \$71 01 DD 01. If no diagnostic messages are sent within 5 seconds after the ECU enters the Bootloader state by using the StayInBoot function, the ECU automatically resets.

在检测到 APP 程序有效后，执行一个 20ms 的延时，如果在延时时间内，收到了 \$31 01 DD 01 的 StayInBoot 的 UDS 诊断报文，ECU 直接进入 Bootloader 的默认会话。在 APP 程

序中强制要求不能支持 0xDD01 的例程 ID。在 BootLoader 里面强制要求支持 0xDD01 的例程 ID，成功进入 Bootloader，ECU 响应 \$71 01 DD 01 的正响应消息。如果 StayInBoot 成功进入 Bootloader 后，5 秒内没有任何诊断报文，ECU 应自动复位。

### 2.2.1.2 Supporting A and B Dual Backup/支持 AB 区双备份

An application in the ECU may not support reprogramming but support dual backup in partitions A and B. Figure 3 shows the flowchart of the Bootloader startup and diagnostic session management of this kind of ECUs.

如果应用程序不支持重编程，并且支持 AB 区双备份的 ECU，其 ECU 的 BootLoader 启动流程图和诊断会话管理参考见 Figure 3 所示。

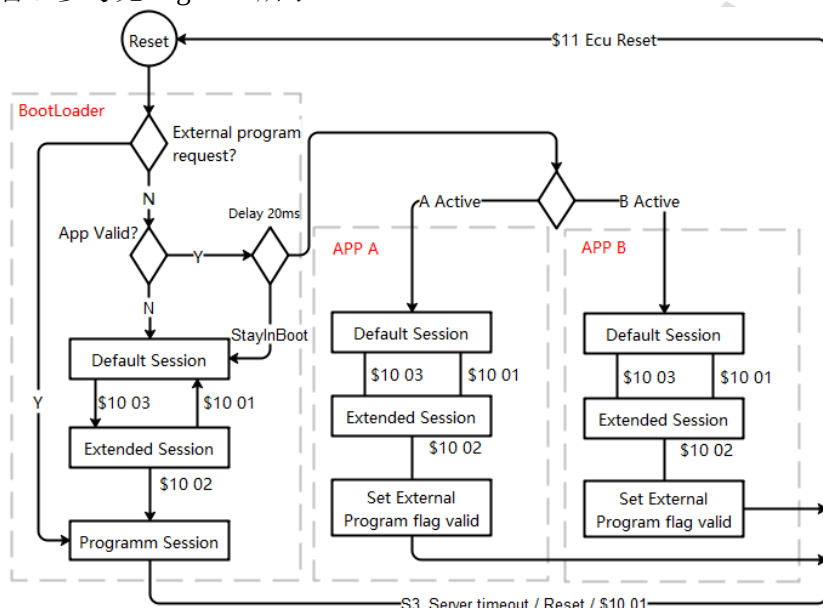


Figure 3: Process of reprogramming not supported and dual backup supported by the application/应用程序不支持重编程但支持 AB 区流程

1. For the ECU whose application does not support reprogramming, it needs to be restarted to switch to the Bootloader for reprogramming after receiving the \$10 02 message. When setting an external programming request, the ECU needs to distinguish whether partition A or B needs to be programmed.

应用程序不支持重编程的 ECU 在接收到 \$10 02 需要重启后切换到 BootLoader 进行重编程，在设置外部编程请求的时候，需要区分本次编程的是 A 区还是 B 区。

2. Requirements for StayInBoot messages are the same as those described in section 2.2.1.1. Details are not described herein.

StayInBoot 报文要求和 2.2.1.1 章节要求相同，在此不再赘述。

### 2.2.2 Application Supporting Reprogramming/应用程序支持编程

For an ECU with an OS (with a file system), the application programming session mode is usually used. Figure 4 shows the flowchart of the Bootloader startup and diagnostic session management of this kind of ECUs.

对智能系统（有文件系统和操作系统）ECU，通常采用在应用程序中支持编程会话，其 ECU 的 BootLoader 启动流程图和诊断会话管理参考见 Figure 4。



- ## 2.3 Security Requirement/安全需求

所有 ECU 必须支持华为智能汽车软件包的安全签名功能，详情请参考《EEA5.0 ECU 软件刷写签名算法规范》。

For intelligent components, follow the specific secure programming solutions (if any). This document is for reference only. For other non-intelligent components, the programming is performed according to this document. The programming of non-intelligent components by intelligent components is compatible with this document.

|                                                 |                  |                  |
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对于智能部件，如果有其他安全刷写解决方案，请按照安全刷写解决方案执行，本规范仅供参考。其他非智能部件，统一遵循本规范接受刷写。智能部件对非智能部件的刷写，保持与本规范兼容。

The storage area of the Bootloader should be protected, including software and hardware protections. For security purposes, whether to enable the StayInBoot function in commercial products depends on the specific project of OEMs.

BootLoader 软件的保存区域应该是受保护的，包括软件保护或者硬件保护。由于安全原因，StayInBoot 功能是否在商用产品中使能，需要根据主机厂具体项目需求来确定。

The input and output of the ECU (if any) should be kept in the security status.

如果是有输入和输出的 ECU，在进行软件刷写过程中应该保证在安全的输入输出状态，防止 ECU 出现非法和不安全的状态。

## 2.4 Other Requirements/其他要求

During ECU programming, the Tester periodically sends the TestPresent (\$3E 80) message for functional addressing to ensure that the Tester is online and the ECU is in a non-default session mode. The ECU needs to ensure that the messages do not block or affect normal programming messages, and bypasses the messages. The TestPresent (\$3E 80) message should not interrupt or affect the execution of any other services of the ECU.

在 ECU 刷写过程中，Tester 周期性发送功能寻址的 TestPresent 消息（\$3E 80）来确保 Tester 在线以及保持 ECU 处于非默认会话模式，ECU 需要确保此消息不要阻塞和影响正常的刷写消息，需要对此消息进行旁路处理。TestPresent 消息（\$3E 80）消息不应该打断或者影响 ECU 的其他任何服务的执行。

If the update fails due to signature verification, programming dependency check, version installation or other causes, the ECU needs to delete data downloaded to the Random Access Memory (RAM) and set the current version to invalid, or deletes the downloaded version.

如果由于签名校验，编程依赖检查，版本安装等原因导致刷写没有成功，ECU 需要删除已经下载到 RAM 中的数据并设置当前版本无效或删除已下载版本。

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### 3 Diagnostic Service/诊断服务

#### 3.1 Service Definitions/服务定义

The following diagnostic services and sub-functions should be supported:

应支持以下诊断服务和子功能:

| Service   | Phys. Req. | Function Req. | SPRMI B | APP  |      |      | Bootloader |      |      | Security |         |         |          | Cvt |
|-----------|------------|---------------|---------|------|------|------|------------|------|------|----------|---------|---------|----------|-----|
|           |            |               |         | Def. | Pro. | Ext. | Def.       | Pro. | Ext. | Loc.     | UnL. L1 | UnL. L2 | UnL. FBL |     |
| \$22 F184 | x          | —             | —       | x    | x    | x    | x          | x    | x    | x        | x       | o       | x        | M   |
| \$22 F189 | x          | —             | —       | x    | x    | x    | —          | —    | —    | x        | x       | o       | —        | M   |
| \$22 F18A | x          | —             | —       | x    | x    | x    | x          | x    | x    | x        | x       | o       | x        | M   |
| \$22 F180 | x          | —             | —       | x    | x    | x    | x          | x    | x    | x        | x       | o       | x        | M   |
| \$22 F187 | x          | —             | —       | x    | x    | x    | x          | x    | x    | x        | x       | o       | x        | M   |
| \$22 F089 | x          | —             | —       | x    | x    | x    | x          | x    | x    | x        | x       | o       | x        | M   |
| \$22 F0F0 | x          | —             | —       | x    | x    | x    | x          | x    | x    | x        | x       | o       | x        | M   |
| \$22 F0F1 | x          | —             | —       | x    | x    | x    | x          | x    | x    | x        | x       | o       | x        | M   |
| \$22 F0F2 | x          | —             | —       | x    | x    | x    | x          | x    | x    | x        | x       | o       | x        | M   |
| \$22 F0F3 | x          | —             | —       | x    | x    | x    | x          | x    | x    | x        | x       | o       | x        | M   |
| \$2E F184 | x          | —             | —       | —    | x    | —    | —          | x    | —    | —        | —       | —       | x        | M   |
| \$31 0203 | x          | —             | —       | —    | —    | x    | —          | —    | x    | x        | —       | —       | —        | M   |

|               |                                                                                                                                                                                                                                                  |   |   |   |   |   |   |   |   |   |   |   |   |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|---|---|
| \$31<br>FF00  | x                                                                                                                                                                                                                                                | - | - | - | o | - | - | x | - | - | - | x | M |
| \$31<br>FF01  | x                                                                                                                                                                                                                                                | - | - | - | x | - | - | x | - | - | - | x | M |
| \$31<br>DD01  | x                                                                                                                                                                                                                                                | - | - | - | - | - | x | - | - | x | - | - | M |
| \$31<br>DD02  | x                                                                                                                                                                                                                                                | - | - | - | x | - | - | x | - | - | - | x | M |
| \$31<br>DD03  | x                                                                                                                                                                                                                                                | - | - | - | x | - | - | - | - | - | - | x | M |
| \$31<br>DD04  | x                                                                                                                                                                                                                                                | - | - | - | - | x | - | - | - | - | x | - | M |
| \$31<br>DD0F  | x                                                                                                                                                                                                                                                | - | - | - | x | - | - | x | - | - | - | x | M |
| \$10<br>01    | <p>For details on the support of services/sub-services, please refer to "Q/SQR E8-20-2022 Vehicle Unified diagnostic services (UDS) Standard".</p> <p>服务/子服务的详细支持情况请参考《Q/SQR E8-20-2022 Vehicle Unified diagnostic services (UDS) Standard》。</p> |   |   |   |   |   |   |   |   |   |   |   |   |
| \$10<br>02    |                                                                                                                                                                                                                                                  |   |   |   |   |   |   |   |   |   |   |   |   |
| \$10<br>03    |                                                                                                                                                                                                                                                  |   |   |   |   |   |   |   |   |   |   |   |   |
| \$11          |                                                                                                                                                                                                                                                  |   |   |   |   |   |   |   |   |   |   |   |   |
| \$27<br>01/02 |                                                                                                                                                                                                                                                  |   |   |   |   |   |   |   |   |   |   |   |   |
| \$27<br>11/12 |                                                                                                                                                                                                                                                  |   |   |   |   |   |   |   |   |   |   |   |   |
| \$14          |                                                                                                                                                                                                                                                  |   |   |   |   |   |   |   |   |   |   |   |   |
| \$19          |                                                                                                                                                                                                                                                  |   |   |   |   |   |   |   |   |   |   |   |   |
| \$22          |                                                                                                                                                                                                                                                  |   |   |   |   |   |   |   |   |   |   |   |   |
| \$29          |                                                                                                                                                                                                                                                  |   |   |   |   |   |   |   |   |   |   |   |   |
| \$2A          |                                                                                                                                                                                                                                                  |   |   |   |   |   |   |   |   |   |   |   |   |
| \$2C          |                                                                                                                                                                                                                                                  |   |   |   |   |   |   |   |   |   |   |   |   |
| \$2E          |                                                                                                                                                                                                                                                  |   |   |   |   |   |   |   |   |   |   |   |   |

|      |  |
|------|--|
| \$2F |  |
| \$31 |  |
| \$28 |  |
| \$85 |  |
| \$3E |  |
| \$34 |  |
| \$36 |  |
| \$37 |  |
| \$38 |  |
|      |  |

|  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  |  | <p>Notes/注:</p> <ul style="list-style-type: none"> <li>Row headings indicate different statuses, column headings indicate different services/sub-functions, and the text indicates whether a service/sub-function is supported in each status.<br/>行标题表示不同的状态，列标题表示不同的服务/子功能，正文表示服务/子功能在每个状态中的支持情况;</li> <li>x indicates that the service or sub-function is supported, - indicates that the service or sub-function is not supported, and o indicates that the support depends on specific cases;<br/>“x”为支持，“-”为不支持，“o”为需根据具体功能定义;</li> <li>Phys. Req. indicates physical addressing, and Funct. Req. indicates functional addressing;<br/>Phys. Req. 意为物理寻址，Funct. Req. 意为功能寻址;</li> <li>SPRMIB is short for Suppress Positive Response Message Indication Bit;<br/>SPRMIB 意为抑制肯定响应标志位;</li> <li>Def. (Default) indicates default session, Pro. (Programming) indicates programming session, and Ext. (Extended) indicates extended session.<br/>Def. (Default) 意为默认会话; Pro. (Programming) 意为编程会话; Ext. (Extended) 意为扩展会话;</li> </ul> <p>APP Pro. is only supported by</p> |
|--|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  |  | <p>the ECUs which support reprogramming in the application.</p> <p>APP 中的编程会话模式仅用于应用程序中支持重编程的 ECU</p> <ul style="list-style-type: none"> <li>Loc. (Locked) indicates the locked status. UnL.L1 (Unlocked Level 1) indicates the unlocked status of extended level 1. UnL.L2 (Unlocked Level 2) indicates the unlocked status of extended level 2. UnL.FBL (Unlocked Level_FBL) indicates the program-level unlock status. Loc. (Locked)</li> </ul> <p>意为锁定状态；UnL.L1 (Unlocked Level1) 意为扩展级 1 级解锁状态；UnL.L2 (Unlocked Level2) 意为扩展级 2 级解锁状态；UnL.FBL (Unlocked Level_FBL) 意为编程级解锁状态；</p> <ul style="list-style-type: none"> <li>M (Mandatory) indicates that the parameter or service must be supported.</li> </ul> <p>M(Mandatory) 意为参数项或服务必须支持；</p> <ul style="list-style-type: none"> <li>U (User option) indicates that the parameter or service can be supported or not, depending on the specific service application.</li> </ul> <p>U(User option) 意为参数项或服务可以支持或者不支持，取决于具体的服务应用。</p> |
|--|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Note: For details about services and sub-functions listed in the preceding table, see the 《QSQR E8-20-2022 Vehicle Unified diagnostic services (UDS) Standard》. This table lists only the diagnostic services or sub-functions that need to be supported for reprogramming. Other diagnostic services are subject to the diagnostic specifications.

注：上表中服务和子功能的详细信息可在《QSQR E8-20-2022 Vehicle Unified diagnostic services (UDS) Standard》文档中找到，此处仅仅列出重编程需要支持的，其他诊断服务支持情况以诊断规范文档为准；

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## 4 ECU Reprogramming Process/ECU 重编程过程

### 4.1 Overview/概述

This section defines a framework for downloading program data to the non-volatile memory of the ECU. 此部分定义了一个框架，用于将程序的数据下载到非易失性 ECU 存储器中。

The ECU reprogramming process can be separated into the pre-programming (programming preparation), programming (software and data downloading), and post-programming (final processing) steps. The ECU programming in the Bootloader and application model must meet this requirement. If the ECU programming in the Bootloader model is directly started by using the StayInBoot function or other methods and the programming precondition check in the pre-programming step cannot be implemented, stub processing is allowed in the Bootloader code.

重编程过程分为三个步骤，包括预编程步骤（编程准备）、编程步骤（软件和数据下载）和后编程步骤（最终处理）。在 Bootloader 和 APP 内的编程都应该满足此规范要求，如果通过 StayInBoot 或者其他方法直接进入 Bootloader 内编程时，对预编程步骤中的预编程条件检查等无法实现时在 Bootloader 代码中允许打桩处理。

If the programming fails, the Tester needs to send the \$11 01 reset command to reset the target ECU, and send the functional addressing of \$10 81 to make all ECUs in the network return to the default session mode. If the programming fails, the ECU should be able to continue the programming capability and support the Tester's programming retry.

如果刷写过程中失败，Tester 需要发送\$11 01 复位命令对刷写目的 ECU 进行复位，并发送\$10 81 的功能寻址使网络中所有 ECU 回到默认会话模式。如果 ECU 刷写失败，部件应该能够保证继续刷写能力，支持 Tester 的刷写重试。

### 4.2 Pre-Programming Step: Programming Preparation/预编程步骤：编程准备

In the pre-programming step, network preparation, programming condition check, and other operations are executed. Figure 5 shows the pre-programming step.

预编程步骤用于为重编程事件准备网络条件以及检查是否满足编程条件等。预编程步骤的流程图如 Figure 5 所示。

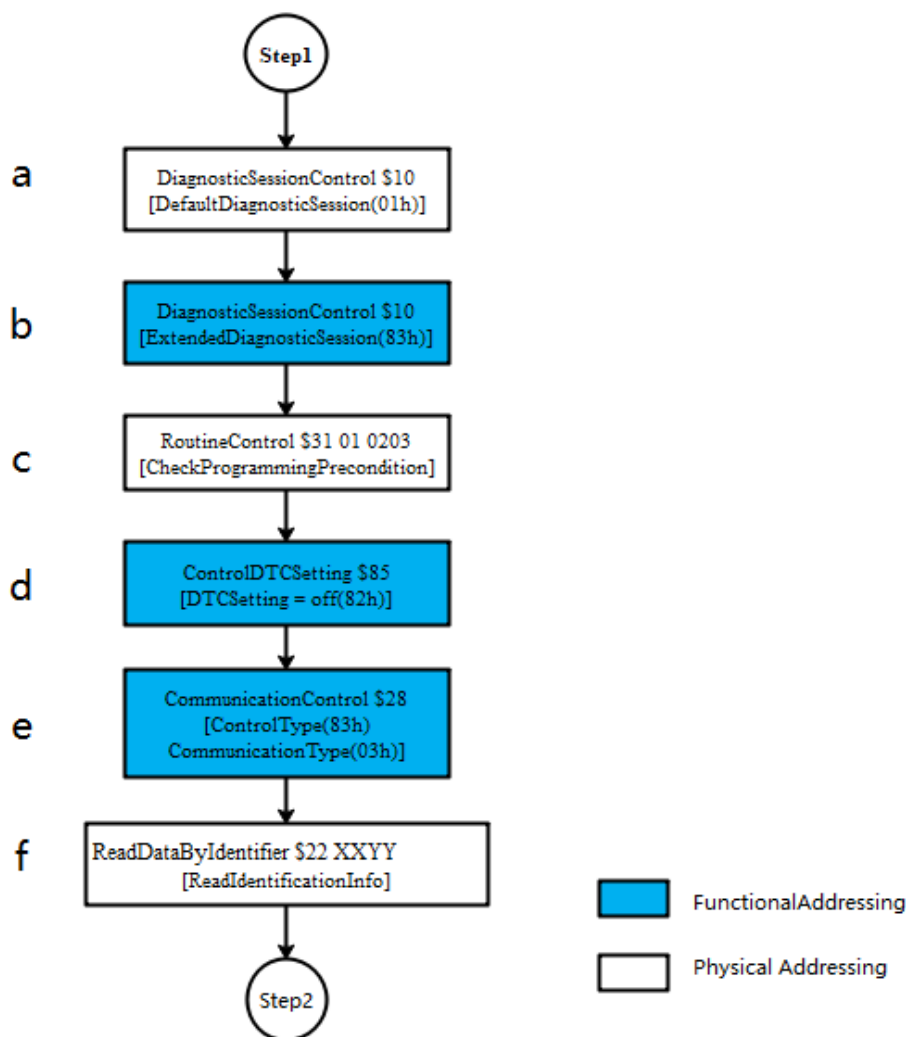


Figure 5: Pre-programming flow/预编程流程图

#### 4.2.1 (a) Diagnostic Session Control/诊断会话控制 (\$10 01)

During the reprogramming process, the physical addressing message of \$10 01 needs to be sent first to request the target ECU to enter the default session mode.

在重编程过程中，首先需要发送\$10 01的物理寻址消息，请求使目的ECU进入默认会话模式。

|              |                          |     |                      |
|--------------|--------------------------|-----|----------------------|
| Direction 方向 | Tester -> ECU, request   |     |                      |
| Service 服务   | DiagnosticSessionControl |     |                      |
| No.序号        | Value 值                  | Cvt | Description 说明       |
| #1           | 10h                      | M   | Service ID 服务 ID     |
| #2           | 01h                      | M   | Default session 默认会话 |
|              |                          |     |                      |

|              |                          |     |                         |
|--------------|--------------------------|-----|-------------------------|
| Direction 方向 | ECU -> Tester, Response  |     |                         |
| Service 服务   | DiagnosticSessionControl |     |                         |
| No.序号        | Value 值                  | Cvt | Description 说明          |
| #1           | 50h                      | M   | Service ID 服务 ID        |
| #2           | 01h                      | M   | Default session 默认会话    |
| #3           | 00-FFh                   | M   | P2ServerMax(HightByte)  |
| #4           | 00-FFh                   | M   | P2ServerMax(LowByte)    |
| #5           | 00-FFh                   | M   | P2*ServerMax(HightByte) |
| #6           | 00-FFh                   | M   | P2*ServerMax(LowByte)   |

#### 4.2.2 (b) Diagnostic Session Control /诊断会话控制 (\$10 83)

During the reprogramming process, a functional addressing message of \$10 83 needs to be sent first to request all ECUs in the network to enter the extended session mode. After sending the message, the Tester waits for 1s and then performs the subsequent operations.

在重编程过程中，首先需要发送\$10 83的功能寻址消息，请求使网络中所有 ECU 进入扩展会话模式。Tester 发送完此报文后，需要延时 1 秒再执行后续步骤。

|              |                          |     |                       |
|--------------|--------------------------|-----|-----------------------|
| Direction 方向 | Tester -> ECU, request   |     |                       |
| Service 服务   | DiagnosticSessionControl |     |                       |
| No.序号        | Value 值                  | Cvt | Description 说明        |
| #1           | 10h                      | M   | Service ID 服务 ID      |
| #2           | 83h                      | M   | Extended session 扩展模式 |

After sending this message, the Tester needs to send the TesterPresent (\$3E 80) service offering functional addressing at an interval of 2s to keep the diagnostic session active until the entire reprogramming process is complete.

发送完此报文后，Tester 需要按照周期为 2 秒的时间，发送功能寻址的 TesterPresent (\$3E 80)服务，以保持诊断会话处于活动状态直到整个重编程流程完成，才能停止发送。

#### 4.2.3 (c) Programming Condition Check/编程条件检查 (\$31 01 0203)

Before the ECU reprogramming, the reprogramming conditions of the system are checked. Ensure that the system is in a secure executable state. For example, the ECU voltage is within the normal range (6 V to 18 V), the vehicle velocity is less than or equal to 5 km/h, the engine is not running, and the gear is in P mode. The to-be-checked pre-programming conditions of each ECU are described in the ECU diagnostic questionnaire. The conditions are agreed by suppliers and OEMs.

在进行重编程前，需要检查系统的重编程条件是否满足。确保系统处于可以实施的安全状态，例如：ECU 电压在正常范围（6V-18V）、车速<=5km/h、发动机未运行、档位处于 P 档。每个 ECU 的具体预编程检查条件在 ECU 的诊断调查表中体现，由供应商和主机厂达成一致。

In a specific programming scenario, if the current controller cannot receive the signal value of the valid judging condition due to communication control (see section 4.2.5), the ECU should check whether the reprogramming condition is met based on the latest valid value received. If the ECU enters the BootLoader through StayInBoot or because the application flag is invalid, the ECU needs to respond to the routine status = 00h to the Tester during the pre-programmed condition check to ensure the normal programming of such scenarios.

在特定刷写场景下如果由于通讯控制（见 4.2.5 章节）引起关联条件信号关闭，导致当前控制器无法接收有效判断条件的信号值时，该控制器应以最近一次收到的有效值进行检查，判断是否满足重编程条件。ECU 如果通过 StayInBoot 或者由于应用程序标志位无效进入 BootLoader 时，ECU 需在预编程条件检查时响应例程状态=00h 给 Tester，确保此类场景的正常刷写。

|              |                         |     |                                                                                                                                                                                                                                                                                                                                                          |
|--------------|-------------------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direction 方向 | Tester -> ECU, request  |     |                                                                                                                                                                                                                                                                                                                                                          |
| Service 服务   | RoutineControl          |     |                                                                                                                                                                                                                                                                                                                                                          |
| No.序号        | Value 值                 | Cvt | Description 说明                                                                                                                                                                                                                                                                                                                                           |
| #1           | 31h                     | M   | Service ID 服务 ID                                                                                                                                                                                                                                                                                                                                         |
| #2           | 01h                     | M   | Start to check 启动检查                                                                                                                                                                                                                                                                                                                                      |
| #3           | 02h                     | M   | 0x0203 programming condition check                                                                                                                                                                                                                                                                                                                       |
| #4           | 03h                     | M   | 0x0203 编程条件检查                                                                                                                                                                                                                                                                                                                                            |
|              |                         |     |                                                                                                                                                                                                                                                                                                                                                          |
| Direction 方向 | ECU -> Tester, Response |     |                                                                                                                                                                                                                                                                                                                                                          |
| Service 服务   | RoutineControl          |     |                                                                                                                                                                                                                                                                                                                                                          |
| No.序号        | Value 值                 | Cvt | Description 说明                                                                                                                                                                                                                                                                                                                                           |
| #1           | 71h                     | M   | Service ID 服务 ID                                                                                                                                                                                                                                                                                                                                         |
| #2           | 01h                     | M   | Start to check 启动检查                                                                                                                                                                                                                                                                                                                                      |
| #3           | 02h                     | M   | 0x0203 programming condition check                                                                                                                                                                                                                                                                                                                       |
| #4           | 03h                     | M   | 0x0203 编程条件检查                                                                                                                                                                                                                                                                                                                                            |
| #5           | 00-FFh                  | M   | 00: indicates that the check is successful 成功<br>01: indicates that other failure causes 其他失败<br>02: indicates that the voltage is abnormal 电压异常<br>03: indicates that the vehicle speed is abnormal 车速异常<br>04: indicates that an engine (or motor) running exception occurs 发动机（或电机）运行异常<br>05: indicates that the gear is not in the P state 挡位未在 P 档 |

|  |  |  |                                                                                                                                                                                                                                      |
|--|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  |  | 06/07/.../FF: indicates a customized failure, respectively. The specified error code shall be associated with the defined programming conditions and reflected in the diagnostic questionnaire 自定义失败原因, 具体错误码需要和定义的编程条件做关联并在诊断调查表中反馈 |
|--|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### 4.2.4 (d) DTC Setting Control/控制 DTC 设置 (\$85)

The Tester uses this message to disable DTC settings of all ECUs to ensure that all ECUs do not record DTCs during reprogramming.

Tester 使用此服务禁用所有 ECU 的 DTC 设置, 以确保所有 ECU 在重编程过程中不会记录 DTC。

|              |                        |     |                                  |
|--------------|------------------------|-----|----------------------------------|
| Direction 方向 | Tester -> ECU, request |     |                                  |
| Service 服务   | ControlDTCSetting      |     |                                  |
| No.序号        | Value 值                | Cvt | Description 说明                   |
| #1           | 85h                    | M   | Service ID 服务 ID                 |
| #2           | 82h                    | M   | Disable the DTC record 关闭 DTC 记录 |

#### 4.2.5 (e) Communication Control/通讯控制 (\$28)

The Tester uses this message to disable DTC settings of all ECUs to ensure that all ECUs do not record DTCs during reprogramming.

Tester 使用此服务禁用所有 ECU 的网络管理报文和通用报文的发送和接收。

|              |                        |     |                                                        |
|--------------|------------------------|-----|--------------------------------------------------------|
| Direction 方向 | Tester -> ECU, request |     |                                                        |
| Service 服务   | CommunicationControl   |     |                                                        |
| No.序号        | Value 值                | Cvt | Description 说明                                         |
| #1           | 28h                    | M   | Service ID 服务 ID                                       |
| #2           | 83h                    | M   | Disable communication 禁止通讯                             |
| #3           | 03h                    | M   | Network management and general messages<br>网络管理报文和通用报文 |

#### 4.2.6 (f) Read Data By Identifier/读取数据 (\$22)

This operation is used to obtain ECU information, such as software fingerprint information, part number, vendor code, hardware version number, software version number, ECU serial number, and Bootloader version number and so on.

此步骤主要用于获取 ECU 的相关信息, 比如读取软件指纹信息、硬件版本号、软件版本号 BootLoader 版本号等信息。

The external diagnostic equipment Tester always reads this information. The execution results of this operation (for example, negative responses from the ECU, no response and timeout of the ECU, or the non-execution of this operation by the Tester) should not affect the subsequent programming process.

外部诊断设备 Tester 总会读取这些信息或者更多的信息，此步骤 ECU 的实现情况不应该影响整个刷写流程。ECU 和 Tester 需要保证此步骤的执行结果（ECU 否定响应，ECU 超时未响应或者 Tester 没有执行此步骤等）不应该影响后续刷写流程的执行。

#### 4.3 Programming Step: Software and Data Downloading/编程步骤：软件和数据下载

In the programming step, data is downloaded to the target ECU that needs to be reprogrammed. This is performed in the Bootloader or an application. The request messages in this step are all physical addressing messages. Figure 6 shows the programming procedure.

编程步骤用于将数据下载到需要重编程的目标 ECU 中。此步骤在 Bootloader 程序或应用程序中执行。此步骤的请求消息类型全部是物理寻址。编程步骤的流程 Figure 6 所示。

|                                                 |                  |                  |
|-------------------------------------------------|------------------|------------------|
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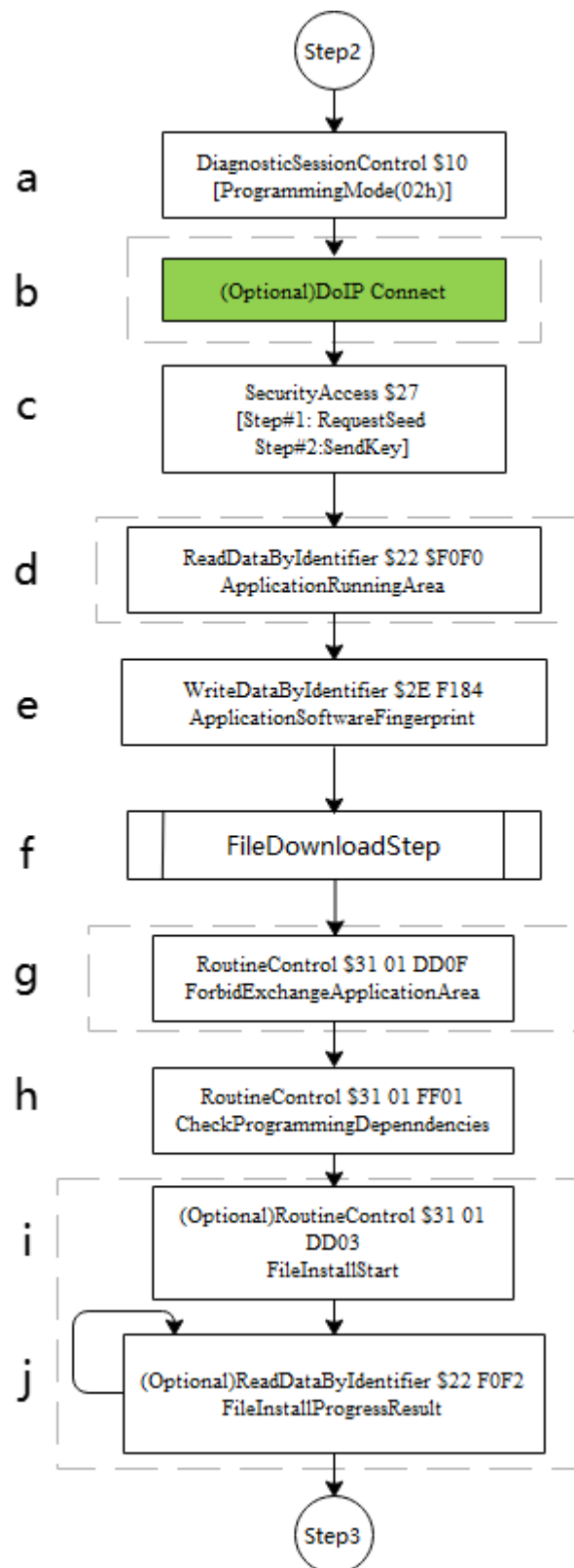


Figure 6: Programming procedure/编程步骤流程图

#### 4.3.1 (a) Diagnostic Session Control/诊断会话控制 (\$10 02)

DiagnosticSessionControl \$10 02 enables the ECU to enter the programming session mode and allocate necessary resources for reprogramming. Before entering the programming session mode, the ECU needs to recheck that the programming conditions (4.2.3 (c)) meet programming requirements. The ECU can exit the programming session mode only through the ECUReset (\$11) service, the DiagnosticSessionControl (\$10 01) service, or S3\_Timeout.

诊断会话控制使 ECU 应进入编程会话模式，并分配重编程所需的所必要资源。在进入编程会话模式前，ECU 内部需要再次检查编程条件（4.2.3 步骤 c），保证进入编程会话时肯定满足编程条件。要离开编程会话模式，只能通过 ECUReset（\$11）服务、诊断会话控制（\$10 01）服务，或 ECU 的 S3\_Timeout 来离开。

If the ECU needs to be restarted to enter the programming session mode, after receiving the \$10 02 message, the ECU returns the NRC 78 message. After resetting and entering the programming session mode, the ECU responds with the positive response message of \$10 02. The ECU to be restarted after entering the programming session mode must reset after actively disabling and clearing the DoIP link. ECU 进入编程会话，如果 ECU 需要重启才能进入编程会话，在收到此消息后，先返回 NRC 78 的消息，待 ECU 复位进入编程会话后响应\$10 02 的正响应。进入编程会话需要重启的 ECU 应该主动关闭和清理 DoIP 链路后再复位。

After sending the message, the Tester waits for 1s and then performs the subsequent operations.  
Tester 发送完此报文后，需要延时 1 秒再执行后续步骤。

|              |                          |     |                          |
|--------------|--------------------------|-----|--------------------------|
| Direction 方向 | Tester -> ECU, request   |     |                          |
| Service 服务   | DiagnosticSessionControl |     |                          |
| No.序号        | Value 值                  | Cvt | Description 说明           |
| #1           | 10h                      | M   | Service ID 服务 ID         |
| #2           | 02h                      | M   | Programming session 编程会话 |
|              |                          |     |                          |
| Direction 方向 | ECU -> Tester, Response  |     |                          |
| Service 服务   | DiagnosticSessionControl |     |                          |
| No.序号        | Value 值                  | Cvt | Description 说明           |
| #1           | 50h                      | M   | Service ID 服务 ID         |
| #2           | 02h                      | M   | Programming session 编程会话 |
| #3           | 00-FFh                   | M   | P2ServerMax(HightByte)   |
| #4           | 00-FFh                   | M   | P2ServerMax(LowByte)     |
| #5           | 00-FFh                   | M   | P2*ServerMax(HightByte)  |
| #6           | 00-FFh                   | M   | P2*ServerMax(LowByte)    |

#### 4.3.2 (b) DoIP Connection Setting Up/建立 DoIP 连接

(Optional) If the ECU reset makes its DoIP link disconnected, the Tester needs to detect and reconnect the link. The Tester delays for 1s and then sends groupcast DoIP Vehicle Discovery Request messages per 500 ms to check whether the DoIP channel between the target ECU and Tester is ready. After receiving the preceding messages, the ECU responds. The Tester attempts to restore the DoIP link to the target ECU after receiving two consecutive correct responses from the target ECU. The default connection timeout period T1 is 10 seconds. When the target ECU correctly responds to DoIP Vehicle Discovery Request messages for two consecutive times or the T1 timer expires, the Tester stops sending DoIP Vehicle Discovery Request messages.

此步骤为可选操作，如果 ECU 复位会导致自己的 DoIP 断链，Tester 需要做链路检测和重连。

Tester 先延时 1 秒，然后周期性发送组播的车辆发现请求消息（DoIP Vehicle Discovery Request 消息）来确认目标 ECU 与 Tester 的 DoIP 通道是否已经就绪，ECU 收到车辆发现请求后给出应答。Tester 以 500ms 的周期发送车辆发现请求，收到目标 ECU 连续 2 次正确响应后，Tester 尝试恢复与目标 ECU 的 DoIP 链路，默认连接超时 T1=10 秒。当目标 ECU 连续 2 次正确响应了车辆发现消息或 T1 定时器超时后，Tester 停止发送车辆发现消息。

#### 4.3.3 (c) Security Access/安全访问 (\$27)

Reprogramming events are sensitive events and must be protected by security authentication. The \$27 11/12 service is used for security authentication. For details about the security authentication process and algorithms, see the 《EEA5.0 ECU 27 算法规范》。

重编程事件为敏感事件，应得到保护，需要进行安全认证解锁。使用\$27 11/12 服务用于安全认证解锁。安全认证解锁具体流程和算法请参考《EEA5.0 ECU 27 算法规范》文档。

#### 4.3.4 (d) Current Running Partition Information Reading/读当前运行分区信息 (\$22 F0F0)

This step must be supported for ECUs that support dual backup. ECUs that do not support dual backup do not support this function. The Tester needs to select the corresponding version file for programming based on the read information about the current running partition. If the ECU returns version A, the Tester will select version B file for programming. If the ECU returns version B, the Tester will select version A file for programming.

对支持双备份的 ECU，必须支持此步。不支持双备份的 ECU 不需要支持。Tester 根据读取到的当前运行分区信息，选择对应的版本文件进行刷写。如果 ECU 返回 A，Tester 将选择 B 版本文件进行刷写，如果 ECU 返回 B，Tester 将选择 A 版本文件进行刷写。

If the ECU does not support this operation or DID, NRC 31 is returned. In this case, the Tester uses the general version file for programming. Otherwise, the Tester uses the version file in partition A or B for programming.

如果 ECU 不支持此步骤，返回 NRC 31 表示不支持此 DID，Tester 取通用版本文件进行刷写，否则应该取对应的 A 或者 B 区的版本文件进行刷写。

|              |                        |     |                |
|--------------|------------------------|-----|----------------|
| Direction 方向 | Tester -> ECU, request |     |                |
| Service 服务   | ReadDataByIdentifier   |     |                |
| No.序号        | Value 值                | Cvt | Description 说明 |

|              |                         |     |                                              |
|--------------|-------------------------|-----|----------------------------------------------|
| #1           | 22h                     | M   | Service ID 服务 ID                             |
| #2           | F0h                     | M   | 0xF0F0 current running partition information |
| #3           | F0h                     | M   | 0xF0F0 当前运行分区信息                              |
|              |                         |     |                                              |
| Direction 方向 | ECU -> Tester, Response |     |                                              |
| Service 服务   | ReadDataByIdentifier    |     |                                              |
| No.序号        | Value 值                 | Cvt | Description 说明                               |
| #1           | 62h                     | M   | Service ID 服务 ID                             |
| #2           | F0h                     | M   | 0xF0F0 current running partition information |
| #3           | F0h                     | M   | 0xF0F0 当前运行分区信息                              |
| #4           | 41/42h                  | M   | "A/B"                                        |

#### 4.3.5 (e) Fingerprint Data Writing/写入指纹 (\$2E F184)

Software fingerprint information, including software modification time and "modified by". The information is written to the flash through \$2E F184. The ECU needs to maintain a counter for the number of programming attempts. Before a positive response is returned, the counter value is incremented by one for each attempt, and should be stored persistently. If the number of programming attempts exceeds the set value, the ECU returns **NRC22**. If the set value is **0**, the number of programming attempts is not limited. The default value is **0**.

软件指纹信息，包含了修改软件的时间，以及谁修改了软件等相关信息，通过\$2E F184 写入 Flash 中保存。ECU 需要维护一个尝试编程次数的计数器 (DID= 0xF0F1)，在正响应返回之前，此计数器加 1，此计数值应该持久保存。如果编程次数超过 ECU 最大尝试编程次数（设置为 0 时，不限制尝试编程次数，默认为 0），ECU 应该返回 NRC22。

|              |                        |     |                                                                                                                                                                                                                |
|--------------|------------------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direction 方向 | Tester -> ECU, request |     |                                                                                                                                                                                                                |
| Service 服务   | WriteDataByIdentifier  |     |                                                                                                                                                                                                                |
| No.序号        | Value 值                | Cvt | Description 说明                                                                                                                                                                                                 |
| #1           | 2Eh                    | M   | Service ID 服务 ID                                                                                                                                                                                               |
| #2           | F1h                    | M   | 0xF184 software fingerprint information                                                                                                                                                                        |
| #3           | 84h                    | M   | 0xF184 软件指纹信息                                                                                                                                                                                                  |
| #4           | 00-FFh                 | M   | Programming time in year. The value ranges from 0 to 255. The value <b>0</b> indicates the year 2000, the value <b>1</b> indicates the year 2001, and so on.<br>刷写时间-年，取值范围 0-255，0 表示 2000 年，1 表示 2001 年以此类推。 |

|              |                         |     |                                                                                                                                                                                                                                                                             |
|--------------|-------------------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| #5           | 00-FFh                  | M   | Programming time in month. The value ranges from 1 to 12.<br>刷写时间-月，取值范围 1 到 12。                                                                                                                                                                                            |
| #6           | 00-FFh                  | M   | Programming time in day. The value ranges from 1 to 31.<br>刷写时间-日，取值范围 1-31。                                                                                                                                                                                                |
| #7           | 00-FFh                  | M   | Serial number of the diagnostic instrument in the format of ASCII characters<br>ASCII 字符，表示诊断仪序号等信息。<br>For over-the-air (OTA) programming, set this parameter to <b>OTA001</b> .<br>如果是 OTA 刷写，填 OTA001。<br>Set this parameter to 0x20 for unused bytes.<br>未使用的字节填写 0x20。 |
| ...          | 00-FFh                  | M   |                                                                                                                                                                                                                                                                             |
| #22          | 00-FFh                  | M   |                                                                                                                                                                                                                                                                             |
|              |                         |     |                                                                                                                                                                                                                                                                             |
| Direction 方向 | ECU -> Tester, Response |     |                                                                                                                                                                                                                                                                             |
| Service 服务   | WriteDataByIdentifier   |     |                                                                                                                                                                                                                                                                             |
| No.序号        | Value 值                 | Cvt | Description 说明                                                                                                                                                                                                                                                              |
| #1           | 6Eh                     | M   | Service ID 服务 ID                                                                                                                                                                                                                                                            |
| #2           | F1h                     | M   | 0xF184 software fingerprint information                                                                                                                                                                                                                                     |
| #3           | 84h                     | M   | 0xF184 软件指纹信息                                                                                                                                                                                                                                                               |

Number of programming attempts counter DID=0xF0F1, the following table describes the format definition:

尝试编程计数器 DID=0xF0F1 的格式定义如下表所示：

|                                                                      |                                                |        |                                                                            |
|----------------------------------------------------------------------|------------------------------------------------|--------|----------------------------------------------------------------------------|
| Number of programming attempts counter<br>0xF0F1<br>尝试编程次数<br>0xF0F1 | Type: unsigned 32-bit integer<br>类型：无符号 32 位整型 | 00-FFh | Number of programming attempts counter Byte#1 (MSB)<br>尝试编程次数 Byte#1 (MSB) |
|                                                                      |                                                | 00-FFh | Number of programming attempts counter Byte#2<br>尝试编程次数 Byte#2             |
|                                                                      |                                                | 00-FFh | Number of programming attempts counter Byte#3<br>尝试编程次数 Byte#3             |
|                                                                      |                                                | 00-FFh | Number of programming attempts counter Byte#4 (LSB)<br>尝试编程次数 Byte#4 (LSB) |

#### 4.3.6 (f) File Downloading/文件下载流程

Traditional components use the service whose SID is 0x34 for programming. For details about how to transfer files, see section 4.5.1.

传统部件使用 SID=0x34 进行刷写，请参考 4.5.1 章节，完成文件的传输。

Components without driver files use the service whose SID is 0x38 for programming. For details about how to transfer files, see section 4.5.2.

对于没有驱动文件的部件使用 SID=0x38 进行刷写，请参考 4.5.2 章节，完成文件的传输。

#### 4.3.7 (g) Automatic Partition Switchover Prevention /阻止自动切区(\$31 01 DD0F)

This step is used to restrict the ECUs in partition A or B to automatically switch to and activate the other partition. Components that support running in partitions A and B must support this step (if they do not support this step, apply to the OEM for approval). Components that do not support running in partitions A and B do not support this step. Traditional ECUs that support partitions A and B can deliver this command to prevent automatic partition switchover after the programming program is transmitted, preventing unexpected automatic partition switchover (For details about the programming scenario of dual backup ECUs, see section 4.6). This step is optional. It can be used in the remote parallel upgrade scenario. In the local upgrade scenario, you can choose not to support this step.

此步骤用于限制支持 A/B 区 ECU 做自动切区激活的动作，支持 A/B 区运行的部件必须支持此步骤（如无法支持需要向 OEM 申请认可），或非 A/B 区运行部件可不支持此步骤。支持 A/B 区的传统 ECU 在刷写程序完成传输后可下发该指令阻止自动切区的动作，防止发生非预期的自动切区动作（双备份控制器刷写场景可以参考 4.6 章节）。此例程为可选步骤，该例程可提供给远程并行升级场景使用，近端升级场景下可以选择不支持。

Once the command for disabling automatic partition switchover is delivered, the ECU needs to disable the automatic update of the boot partition flag in the next step that contains the automatic partition switchover activation action (for example, the dependency check step in 4.3.8(h) Programming Dependency Check/检查编程依赖关系 (\$31 01 FF01) or the reset step in 4.4.2(b) ECU Reset/复位 (\$11)) in the current power-on period. After the automatic update of the boot partition flag is disabled, the function of preventing automatic partition switchover must be disabled. During the running partition switchover process (see section 4.6.2), the ECU should forcibly switch between partitions A and B immediately and ignore the command for preventing automatic partition switchover. Section 4.6.3 in this document provides the implementation solution for preventing automatic partition switchover for reference. 一旦下发启动阻止自动切区指令，ECU 需要在本次上电周期内的下一次包含自动切区激活动作的步骤（如 4.3.8 章节依赖性检查步骤或者 4.4.2 章节复位步骤）中禁止自动更新启动分区标识，在完成禁止自动更新启动分区标识后，阻止自动切区即须失效。在运行分区切换流程中（参考 4.6.2 章节），ECU 应该强制立即做 A/B 区切换并忽略阻止自动切区命令。本规范 4.6.3 章节提供了阻止自动切区的实现方案供参考。

|              |                        |     |                  |
|--------------|------------------------|-----|------------------|
| Direction 方向 | Tester -> ECU, request |     |                  |
| Service 服务   | RoutineControl         |     |                  |
| No.序号        | Value 值                | Cvt | Description 说明   |
| #1           | 31h                    | M   | Service ID 服务 ID |

|              |                        |     |                                                                                                                                                          |
|--------------|------------------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| #2           | 01h                    | M   | Start 启动                                                                                                                                                 |
| #3           | DDh                    | M   | 0xDD0F automatic partition switchover prevention                                                                                                         |
| #4           | 0Fh                    | M   | 0xDD0F 阻止切区动作                                                                                                                                            |
|              |                        |     |                                                                                                                                                          |
| Direction 方向 | Tester -> ECU, request |     |                                                                                                                                                          |
| Service 服务   | RoutineControl         |     |                                                                                                                                                          |
| No.序号        | Value 值                | Cvt | Description 说明                                                                                                                                           |
| #1           | 71h                    | M   | Service ID 服务 ID                                                                                                                                         |
| #2           | 01h                    | M   | Start 启动                                                                                                                                                 |
| #3           | DDh                    | M   | 0xDD0F automatic partition switchover prevention                                                                                                         |
| #4           | 0Fh                    | M   | 0xDD0F 阻止切区动作                                                                                                                                            |
| #5           | 00/01h                 | M   | 00: indicates that the check is successful 成功<br>01: indicates that the check fails 失败 (Note: Provide any failure cause in the diagnosis questionnaire.) |

#### 4.3.8 (h) Programming Dependency Check/检查编程依赖关系 (\$31 01 FF01)

The RoutineControl \$31 01 FF01 message is sent to trigger the ECU to check the programming dependencies when software and data of all applications in the ECU is downloaded. The ECU should internally checks whether the downloaded file version is compatible with the hardware version, whether the software version depends on other ECU versions, whether the software version is compatible with configuration files, and whether the software version is compatible with the calibration data. This document does not provide mandatory specifications for specific checks. The ECU only needs to perform the compatibility check based on its own requirements. This operation is mandatory. If the ECU does not need to check compatibilities and dependencies, a success message is returned.

当 ECU 的所有应用程序软件/数据都已全部下载完毕，发送此消息，触发 ECU 进行编程依赖关系检查。建议 ECU 内部对下载文件版本和硬件版本是否兼容检查，软件版本是否依赖其他 ECU 版本，软件版本和配置文件是否兼容，软件版本和标定数据是否兼容等，本文档不对具体检查做强制规范，ECU 根据自己的要求做兼容检查即可。如果 ECU 确定不需要检查兼容性和依赖性，那么直接返回成功即可。

The ECU maintains a programming dependency check succeed counter and saves it in the flash. The count value is incremented by one each time the dependency check is successful. When the maximum number of programming attempts is reached (if the maximum number of programming attempts is set to 0, the number of programming times is not limited, and the default value is 0), the ECU should disable programming. After the programming dependency check, the ECU returns the negative response message of NRC22.

ECU 内部维护一个编程依赖检查成功的计数器 (DID= 0xF0F3)，并保存在 Flash 中，每次编程依赖检查成功后此计数器加 1，当达到最大允许编程次数（最大允许编程次数设置为 0 时，不限制编程次数，默认为 0），ECU 应该禁止编程，编程依赖检查返回 NRC22 的否定响应消息。

|              |                         |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------|-------------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direction 方向 | Tester -> ECU, request  |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Service 服务   | RoutineControl          |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| No.序号        | Value 值                 | Cvt | Description 说明                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| #1           | 31h                     | M   | Service ID 服务 ID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| #2           | 01h                     | M   | Start 启动                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| #3           | FFh                     | M   | 0xFF01 programming dependency check                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| #4           | 01h                     | M   | 0xFF01 检查编程依赖性                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|              |                         |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Direction 方向 | ECU -> Tester, Response |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Service 服务   | RoutineControl          |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| No.序号        | Value 值                 | Cvt | Description 说明                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| #1           | 71h                     | M   | Service ID 服务 ID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| #2           | 01h                     | M   | Start 启动                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| #3           | FFh                     | M   | 0xFF01 programming dependency check                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| #4           | 01h                     | M   | 0xFF01 检查编程依赖性                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| #5           | 00-FFh                  | M   | 00: indicates that the check is successful 成功<br>01: indicates that other failure causes 其他失败<br>02: indicates that software and hardware are not compatible 软件和硬件不兼容<br>03: indicates that software and configuration files are not compatible 软件和配置文件不兼容<br>04: indicates that software and calibrated data are not compatible 软件和标定数据不兼容<br>05/06/.../FF: indicates a customized failure, respectively. The specified error code shall be associated with the defined programming conditions and reflected in the diagnostic questionnaire 自定义失败原因，具体错误码需要和定义的编程依赖条件做关联并在诊断调查表中反馈 |

Programming dependency check succeed counter DID=0xF0F3, the following table describes the format definition.

编程成功计数器 DID=0xF0F3 的格式定义如下表所示：

|                                                                           |                                                |        |                                                                                       |
|---------------------------------------------------------------------------|------------------------------------------------|--------|---------------------------------------------------------------------------------------|
| programming dependency check succeed counter 0xF0F3<br>编程依赖检查成功计数器 0xF0F3 | Type: unsigned 32-bit integer<br>类型：无符号 32 位整型 | 00-FFh | programming dependency check succeed counter Byte#1 (MSB)<br>编程依赖检查成功计数器 Byte#1 (MSB) |
|                                                                           |                                                | 00-FFh | programming dependency check succeed counter Byte#2<br>编程依赖检查成功计数器 Byte#2             |
|                                                                           |                                                | 00-FFh | programming dependency check succeed counter Byte#3<br>编程依赖检查成功计数器 Byte#3             |
|                                                                           |                                                | 00-FFh | programming dependency check succeed counter Byte#4 (LSB)<br>编程依赖检查成功计数器 Byte#4 (LSB) |

#### 4.3.9 (i) Version Installation Start/开始安装版本 (\$31 01 DD03)

This operation is used to start the ECU version installation process. This operation must be supported by intelligent components, not supported for non-intelligent components. ECU needs to set the program in the partition where the version needs to be installed to invalid before starting the installation. This prevents the application from being executed accidentally when the installation fails. Once the ECU version installation is started, the ECU cannot be called to stop version installation. Wait until the ECU version installation is successful or fails, or the installation timeout occurs.

此步骤用于启动 ECU 的版本安装流程，为可选步骤，智能部件必须支持此步骤，非智能部件不支持此步骤。ECU 需要先把需要安装版本区域的程序设置为无效，再启动安装，防止安装过程没有成功而程序意外执行应用程序。一旦启动 ECU 版本安装，不支持调用停止安装，请等待 ECU 成功完成安装、安装失败或者安装超时。

The ECU must respond to the RoutineControl \$31 01 DD03 message before performing the version installation. If the ECU needs to reset during the version installation, the ECU that supports the DoIP must actively disable the DoIP link first. After executing the message successfully, the Tester waits for 1s and attempts to restore the link to the ECU according to method described in Section 4.3.2 (timeout period T1 is 5 minutes). Then, the Tester performs subsequent operations.

ECU 需要先响应此消息再执行版本安装的流程。如果 ECU 在安装版本过程中需要进行复位，DoIP 的设备必须主动关闭 DoIP 链路后再复位安装。Tester 在此消息执行成功后，延时 1 秒，然后按照 4.3.2 章节的方法尝试恢复和 ECU 的链路（超时时间 T1=5 分钟），再执行后续操作。

|              |                        |     |                  |
|--------------|------------------------|-----|------------------|
| Direction 方向 | Tester -> ECU, request |     |                  |
| Service 服务   | RoutineControl         |     |                  |
| No.序号        | Value 值                | Cvt | Description 说明   |
| #1           | 31h                    | M   | Service ID 服务 ID |

|              |                         |     |                                                      |
|--------------|-------------------------|-----|------------------------------------------------------|
| #2           | 01h                     | M   | Start 启动                                             |
| #3           | DDh                     | M   | 0xDD03 version installation                          |
| #4           | 03h                     | M   | 0xDD03 版本安装                                          |
|              |                         |     |                                                      |
| Direction 方向 | ECU -> Tester, Response |     |                                                      |
| Service 服务   | RoutineControl          |     |                                                      |
| No.序号        | Value 值                 | Cvt | Description 说明                                       |
| #1           | 71h                     | M   | Service ID 服务 ID                                     |
| #2           | 01h                     | M   | Start 启动                                             |
| #3           | DDh                     | M   | 0xDD03 version installation                          |
| #4           | 03h                     | M   | 0xDD03 版本安装                                          |
| #5           | 00/01h                  | M   | 00: indicates success 成功<br>01: indicates failure 失败 |

#### 4.3.10 (j) Installation Progress and Result Obtaining/获取安装进度和结果 (\$22 F0F2)

The Tester should send this message at an interval of 1s to obtain the installation progress and results. This operation must be supported by intelligent components, not supported for non-intelligent components. If the ECU does not respond to this message within the specified period, the Tester cannot regard it as an error and should continue to query until the query is successful and the installation progress reaches 100% or the ECU responds that the installation verification fails. During this process, after detecting that the DoIP link is disconnected, the Tester should reconnect the link until the results that the installation succeeds or fails are obtained. The ECU version may fail to be installed due to faults. In this operation, a timeout period should be set. If the installation is not complete or ends within 30 minutes (timeout period T2), the Tester should terminate the reprogramming process.

Tester 应该以 1 秒的周期调用此服务，获取安装进度和结果。此步骤为可选步骤，智能部件必须支持此步骤，非智能部件不支持此步骤。ECU 超时未响应此消息，Tester 不能当做错误处理，应该继续查询，直到获取到成功并且进度 100%或者 ECU 响应安装或校验失败。此过程中 Tester 如果检测 DoIP 链路断开，Tester 应该做链路重连，直到获取安装成功或者安装失败。ECU 可能由于故障原因一直无法完成安装，此步骤应该设置超时时间，超时时间 T2=30 分钟没有完成或者结束，Tester 应该终止本次重编程流程。

For the response during the installation progress obtaining of the programming step, the value of the FileInstallStatus field can only be 00h, 01h, 02h and 04h.

在编程序列的安装进度获取中，FileInstallStatus 字段应该只允许响应 00h/01h/02h/04h，不允许响应 03h/05h。

|              |                        |
|--------------|------------------------|
| Direction 方向 | Tester -> ECU, request |
|--------------|------------------------|

|              |                         |     |                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------|-------------------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Service 服务   | ReadDataByIdentifier    |     |                                                                                                                                                                                                                                                                                                                                                                                                            |
| No.序号        | Value 值                 | Cvt | Description 说明                                                                                                                                                                                                                                                                                                                                                                                             |
| #1           | 22h                     | M   | Service ID 服务 ID                                                                                                                                                                                                                                                                                                                                                                                           |
| #3           | F0h                     | M   | 0xF0F2 software installation results                                                                                                                                                                                                                                                                                                                                                                       |
| #4           | F2h                     | M   | 0xF0F2 软件安装结果                                                                                                                                                                                                                                                                                                                                                                                              |
|              |                         |     |                                                                                                                                                                                                                                                                                                                                                                                                            |
| Direction 方向 | ECU -> Tester, Response |     |                                                                                                                                                                                                                                                                                                                                                                                                            |
| Service 服务   | ReadDataByIdentifier    |     |                                                                                                                                                                                                                                                                                                                                                                                                            |
| No.序号        | Value 值                 | Cvt | Description 说明                                                                                                                                                                                                                                                                                                                                                                                             |
| #1           | 62h                     | M   | Service ID 服务 ID                                                                                                                                                                                                                                                                                                                                                                                           |
| #3           | F0h                     | M   | 0xF0F2 software installation results                                                                                                                                                                                                                                                                                                                                                                       |
| #4           | F2h                     | M   | 0xF0F2 软件安装结果                                                                                                                                                                                                                                                                                                                                                                                              |
| #5           | 00-05h                  | M   | <b>FileInstallStatus/ FileActiveStatus</b><br><b>00:</b> indicates success.成功<br><b>01:</b> indicates other failures 其他失败<br><b>02:</b> indicates that the version is installing 安装中<br><b>03:</b> indicates that the version is being verified (or activated)校验中（激活中）<br><b>04:</b> indicates that installation fails 安装失败<br><b>05:</b> indicates that the verification (or activation) fails 校验失败（激活失败） |
| #6           | 00-64h                  | M   | Installation progress <b>(verification, or activation)</b> (%). For example, <b>63h</b> indicates that the installation progress is 99%.<br>安装进度 <b>(或校验激活进度)</b> 百分比 (%) 如 63h=99%                                                                                                                                                                                                                        |

#### 4.4 Post-Programming Step: Final Processing/后编程步骤：最终处理

In the post-programming step, final software installation verification results need to be obtained, and the vehicle network and DTC recording function needs to be restored.

后编程步骤用于获取最终的软件安装的校验结果，以及恢复整车网络及 DTC 记录功能。

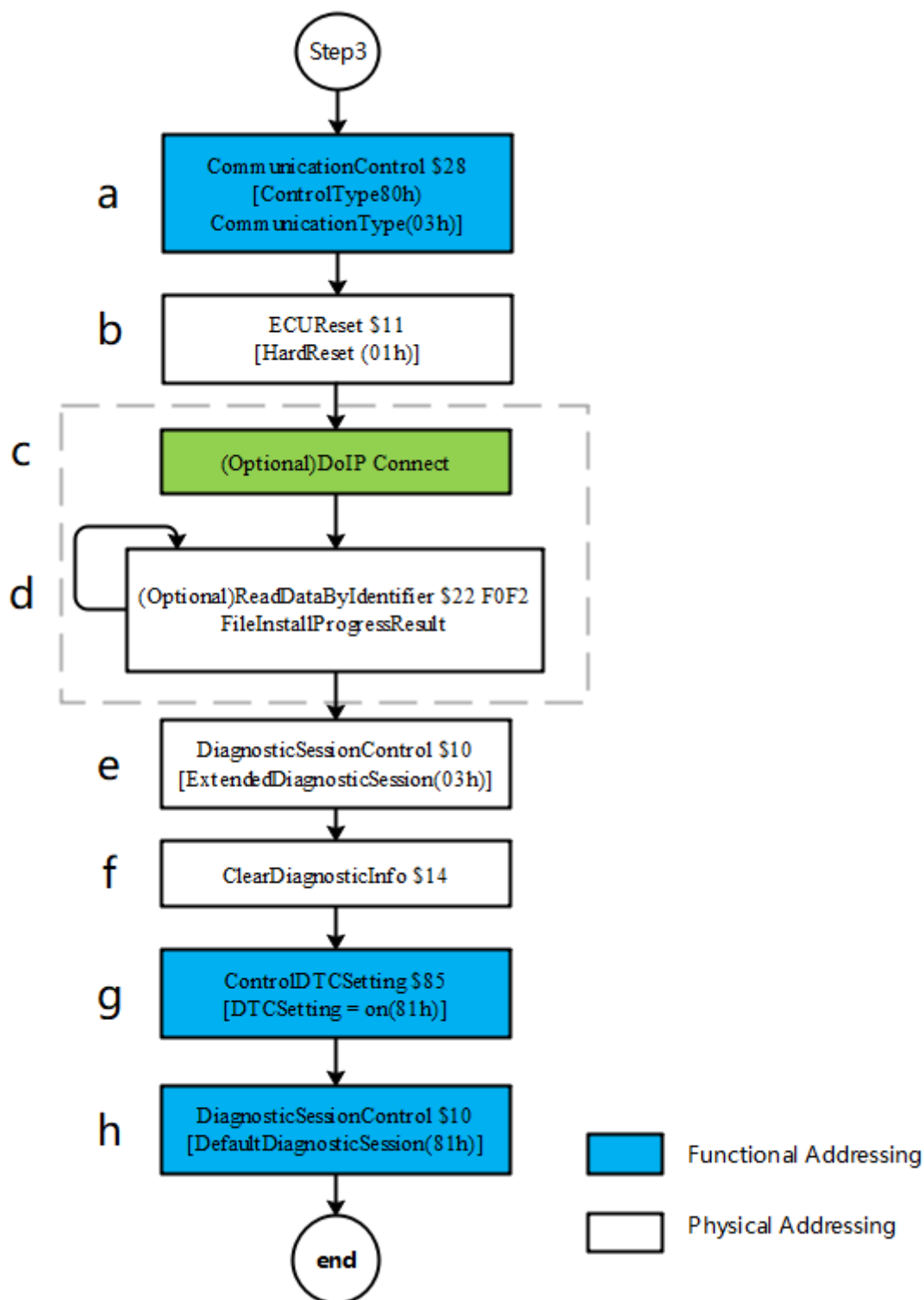


Figure 7: Post-programming procedure/ 后编程步骤流程图

#### 4.4.1 (a) Communication Control/通讯控制 (\$28)

This message is sent to restore the transmission and reception of network management and general messages for all ECUs.

发送此服务恢复所有 ECU 的网络管理报文和通用报文的发送和接收。

|              |                        |     |                                                        |
|--------------|------------------------|-----|--------------------------------------------------------|
| Direction 方向 | Tester -> ECU, request |     |                                                        |
| Service 服务   | CommunicationControl   |     |                                                        |
| No.序号        | Value 值                | Cvt | Description 说明                                         |
| #1           | 28h                    | M   | Service ID 服务 ID                                       |
| #2           | 80h                    | M   | Enable communication 开启通讯                              |
| #3           | 03h                    | M   | Network management and general messages<br>网络管理报文和通用报文 |

#### 4.4.2 (b) ECU Reset/复位 (\$11)

Sending this message will restart the ECU and make the downloaded application software or data take effect. After receiving this message, the ECU actively disables and clears the DOIP link before resetting. After sending the message, the Tester waits for 1s and then performs the subsequent operations.

发送此服务将使 ECU 重启，并生效下载的应用软件/数据。ECU 在收到此消息复位前，应该主动关闭和清理 DoIP 链路。Tester 在发送完此命令后，需要等待 1 秒，再执行后续步骤。

The ECU performs specific initialization operations according to the power-on and startup sequence to implement hard reset. The ECU is not forced to perform a power-off reset.

ECU 的硬复位，指 ECU 按照上电/启动顺序执行特定的初始化操作，不强制 ECU 进行掉电复位。

|              |                         |     |                  |
|--------------|-------------------------|-----|------------------|
| Direction 方向 | Tester -> ECU, request  |     |                  |
| Service 服务   | ECUReset                |     |                  |
| No.序号        | Value 值                 | Cvt | Description 说明   |
| #1           | 11h                     | M   | Service ID 服务 ID |
| #2           | 01h                     | M   | HardRest 硬复位     |
| Direction 方向 | ECU -> Tester, Response |     |                  |
| Service 服务   | ECUReset                |     |                  |
| No.序号        | Value 值                 | Cvt | Description 说明   |
| #1           | 51h                     | M   | Service ID 服务 ID |
| #2           | 01h                     | M   | HardRest 硬复位     |

#### 4.4.3 (c) DoIP Connection Setting Up/建立 DoIP 连接

(Optional). After the ECU that supports the DoIP resets, the diagnostic link will be disconnected. In this case, the Tester needs to detect and reconnect the link. The Tester attempts to restore the link to the ECU according to the method described in section 4.3.2 (timeout period T1 is 5 minutes, this timeout tester can be configured to meet the special reprogramming scenarios of some components). For traditional components, the Tester waits for 5s before performing subsequent operations.

此步骤为可选操作，ECU 复位后，如果是 DoIP 设备，会导致诊断链路断链，在这里需要做链路检测和重连。按照 4.3.2 章节的方法（超时时间 T1=5 分钟，为满足某些部件的特殊刷写场景，此超时时间 Tester 可配置），尝试恢复和 ECU 之间的链路。如果是传统部件，Tester 直接延时 5 秒再进行后续操作。

#### 4.4.4 (d) Post-Installation Verification Result Obtaining/获取安装后校验结果 (\$22 F0F2)

(Optional). This operation must be supported by intelligent components, not supported for non-intelligent components. After an intelligent component completes version installation and restarts, the Tester may need to query the internal installation and verification information to check whether the update is successful. The information includes the update results, failure cause, and error code of the ECU proprietary components. For the response during the installation progress obtaining of the post-programming step, the value of the FileActiveStatus field can only be 00h, 01h, 03h and 05h. For details, see section 4.3.9.

此步骤为可选步骤，智能部件必须支持此步骤，非智能部件不支持此步骤。智能部件完成版本安装并重启后，运行在新版本后可能需要查询内部安装和校验的信息才能确认本次升级是否成功，如 ECU 的私有部件的升级结果，失败原因，故障码等，在后编程序列的激活进度获取中，FileActiveStatus 字段应该只允许响应 00h/01h/03h/05h，不允许响应 02h/04h。此步骤详细信息，请参考 4.3.9 步骤。

#### 4.4.5 (e) Diagnostic Session Control/诊断会话控制 (\$10 03)

The Tester sends the physical addressing message of \$10 03 to request the target ECU to enter the extended session mode.

发送\$10 03的物理寻址消息，请求目的 ECU 进入扩展会话模式。

|              |                          |     |                       |
|--------------|--------------------------|-----|-----------------------|
| Direction 方向 | Tester -> ECU, request   |     |                       |
| Service 服务   | DiagnosticSessionControl |     |                       |
| No.序号        | Value 值                  | Cvt | Description 说明        |
| #1           | 10h                      | M   | Service ID 服务 ID      |
| #2           | 03h                      | M   | Extended session 扩展会话 |
|              |                          |     |                       |
| Direction 方向 | ECU -> Tester, Response  |     |                       |
| Service 服务   | DiagnosticSessionControl |     |                       |
| No.序号        | Value 值                  | Cvt | Description 说明        |

|    |        |   |                         |
|----|--------|---|-------------------------|
| #1 | 50h    | M | Service ID 服务 ID        |
| #2 | 03h    | M | Extended session 扩展会话   |
| #3 | 00-FFh | M | P2ServerMax(HightByte)  |
| #4 | 00-FFh | M | P2ServerMax(LowByte)    |
| #5 | 00-FFh | M | P2*ServerMax(HightByte) |
| #6 | 00-FFh | M | P2*ServerMax(LowByte)   |

#### 4.4.6 (f) Clearing DTC/清除故障码 (\$14)

The Tester clears DTCs of the target ECU by using the physical addressing message.

Tester 使用物理寻址清除目的 ECU 的故障码。

|              |                        |     |                           |
|--------------|------------------------|-----|---------------------------|
| Direction 方向 | Tester -> ECU, request |     |                           |
| Service 服务   | ClearDiagnosticInfo    |     |                           |
| No.序号        | Value 值                | Cvt | Description 说明            |
| #1           | 14h                    | M   | Service ID 服务 ID          |
| #2           | FFh                    | M   | Clear all DTCs<br>清除所有故障码 |
| #3           | FFh                    | M   |                           |
| #4           | FFh                    | M   |                           |

#### 4.4.7 (g) DTC Control Setting/控制 DTC 设置 (\$85)

The Tester uses this message to enable the DTC setting of all ECUs.

Tester 使用此服务使能所有 ECU 的 DTC 设置，启动所有 ECU 的设置 DTC 功能。

|              |                        |     |                                 |
|--------------|------------------------|-----|---------------------------------|
| Direction 方向 | Tester -> ECU, request |     |                                 |
| Service 服务   | ControlDTCSetting      |     |                                 |
| No.序号        | Value 值                | Cvt | Description 说明                  |
| #1           | 85h                    | M   | Service ID 服务 ID                |
| #2           | 81h                    | M   | Enable the DTC record 开启 DTC 记录 |

#### 4.4.8 (h) Diagnostic Session Control/诊断会话控制 (\$10 81)

The Tester will send this message to make all ECUs start the default session mode. After returning to the default session mode, the ECU automatically enables the communication and DTCs.

Tester 将发送此消息使所有 ECU 启动默认会话模式。回到默认会话后，ECU 会自动打开通讯，使能故障码。

|              |                        |
|--------------|------------------------|
| Direction 方向 | Tester -> ECU, request |
|--------------|------------------------|

|            |                          |     |                      |
|------------|--------------------------|-----|----------------------|
| Service 服务 | DiagnosticSessionControl |     |                      |
| No.序号      | Value 值                  | Cvt | Description 说明       |
| #1         | 10h                      | M   | Service ID 服务 ID     |
| #2         | 81h                      | M   | Default session 默认会话 |

## 4.5 Sub-Processes/子流程说明

### 4.5.1 Process of Downloading Traditional Component Files/传统部件文件下载流程

For traditional components, the application storage area may be accidentally erased and rewritten during programming of the application that runs normally. The ECU needs to use a software interlock mechanism, that is, driver code cannot be kept in a permanent memory of the ECU but is downloaded to a RAM of the ECU in the programming process. The files must be downloaded by the sequence of requesting downloading (34h), transmitting data (36h), and requesting to end transmission (37h). After the programming is complete, the driver code saved in the RAM should be cleared. Figure 8 shows the file downloading procedure of the FileDownloadStep. In the step of erasing the memory, if the Tester erases the memory by logical block, it sends the erasure routine only once for complete erasure to download the file. If the Tester erases the memory by software segment, it can send the erasure routines for multiple times according to the address information of multiple software segments in the upgrade file.

传统部件为了防止应用程序正常运行过程中，意外对应用程序存储区进行擦除和重写。ECU 需要采用软件互锁机制，即驱动程序代码不允许保留在 ECU 的永久存储器中，而是在编程过程将驱动程序下载到 ECU 的 RAM 中。下载应遵循指定的顺序，包括请求下载（34h）、传输数据（36h）和请求传输退出（37h）。在刷写完毕后，保存在 RAM 中的驱动程序代码应该清除。FileDownloadStep 的下载流程图见如下 Figure 8 所示。在擦除内存步骤，如果 Tester 是按逻辑块擦除，下载一个文件只发送一次擦除例程进行完整的擦除；如果 Tester 是按软件段擦除，允许按升级文件中多个软件段地址信息发送多次擦除例程。

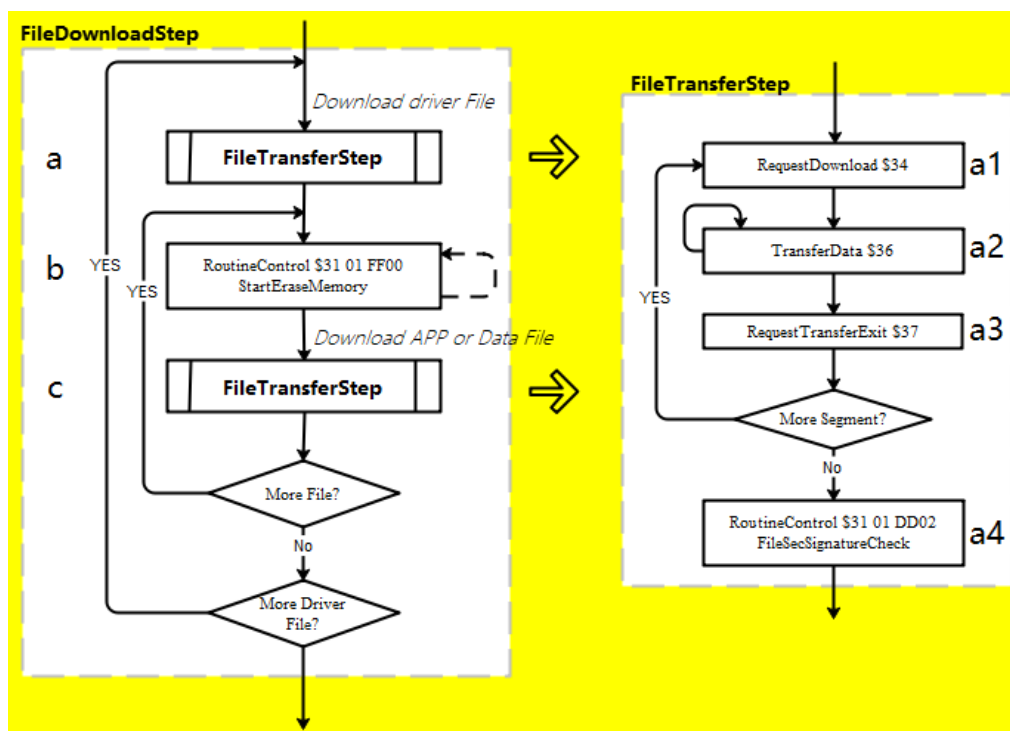


Figure 8: Process of downloading traditional component files/传统部件文件下载流程

#### 4.5.1.1 (a) Downloading Driver Files/驱动文件下载

The step for downloading driver files of a traditional component must be supported.  
传统部件的驱动文件下载步骤是必须支持的。

#### 4.5.1.2 (a1) Requesting to Download Files/请求下载 (\$34)

The Tester sends this message to request the ECU to provide the start address and memory size of data to be programmed in the TransferData \$36 service. The memory address must be consecutive from the start address specified by the RequestDownload service. If DataFormatIdentifier is set to \$10 for compressed files, MemorySize uses an uncompressed value. If the file size exceeds 4 GB, the file is downloaded according to the process of downloading intelligent component files in section 4.5.2.

Tester 使用此请求向 ECU 请求将在 TransferData \$36 服务中编程的数据的起始地址和内存大小。内存位置地址应从 RequestDownload 指定的起始地址开始连续。文件如果压缩 DataFormatIdentifier= \$10, MemorySize 使用未压缩的值。如果文件大小超过 4G, 请使用 4.5.2 章节智能部件文件下载流程进行文件下载。

|              |                        |     |                                                                |
|--------------|------------------------|-----|----------------------------------------------------------------|
| Direction 方向 | Tester -> ECU, request |     |                                                                |
| Service 服务   | RequestDownload        |     |                                                                |
| No.序号        | Value 值                | Cvt | Description 说明                                                 |
| #1           | 34h                    | M   | Service ID 服务 ID                                               |
| #2           | 00h/10h                | M   | 00h: indicates that the file does not need to be compressed or |

|              |                         |     |                                                                                                                             |
|--------------|-------------------------|-----|-----------------------------------------------------------------------------------------------------------------------------|
|              |                         |     | encrypted 不压缩不加密<br>10h: indicates that the file needs to be compressed but not encrypted 压缩不加密                             |
| #3           | 44h                     | M   | addressAndLengthFormatIdentifier<br>bit7-bit4: indicates data length 数据长度<br>bit3-bit0: indicates data start address 数据起始地址 |
| #4           | 00-FFh                  | M   | memoryAddress Byte#1 (MSB)                                                                                                  |
| #5           | 00-FFh                  | M   | memoryAddress Byte#2                                                                                                        |
| #6           | 00-FFh                  | M   | memoryAddress Byte#3                                                                                                        |
| #7           | 00-FFh                  | M   | memoryAddress Byte#4                                                                                                        |
| #8           | 00-FFh                  | M   | memorySize Byte#1 (MSB)                                                                                                     |
| #9           | 00-FFh                  | M   | memorySize Byte#2                                                                                                           |
| #10          | 00-FFh                  | M   | memorySize Byte#3                                                                                                           |
| #11          | 00-FFh                  | M   | memorySize Byte#4                                                                                                           |
|              |                         |     |                                                                                                                             |
| Direction 方向 | ECU -> Tester, Response |     |                                                                                                                             |
| Service 服务   | RequestDownload         |     |                                                                                                                             |
| No.序号        | Value 值                 | Cvt | Description 说明                                                                                                              |
| #1           | 74h                     | M   | Service ID 服务 ID                                                                                                            |
| #2           | 40h                     | M   | Length format identifier 长度格式标识符                                                                                            |
| #3           | 00-FFh                  | M   | maxNumberOfBlockLength Byte#1 (MSB)                                                                                         |
| #4           | 00-FFh                  | M   | maxNumberOfBlockLength Byte#2                                                                                               |
| #5           | 00-FFh                  | M   | maxNumberOfBlockLength Byte#3                                                                                               |
| #6           | 00-FFh                  | M   | maxNumberOfBlockLength Byte#4                                                                                               |

#### 4.5.1.3 (a2) Transmitting Data/传输数据 (\$36)

The Tester uses this message to transmit programmed data to the ECU. The **BlockSequenceCounter** parameter is used as a block counter to ensure that blocks are correctly transmitted. The value of the **BlockSequenceCounter** parameter starts from and increases by one. When the value reaches **FFh**, it starts from **0** next time. During the transmission, the Tester may retransmit the \$36 message. The ECU needs to correctly process the message transmission in this scenario according to the ISO 14229 standard. To prevent infinite transmission of the \$36 message, the ECU counts the number of times that the BlockSequenceCounter parameter is reset. When the number of times reaches the specified value, the negative response message of NRC 73 is returned.

Tester 使用此服务将编程的数据传输到 ECU。参数 BlockSequenceCounter 用作块计数器，以确保 Blocks 正确的被传输。BlockSequenceCounter 从 1 开始，依次累加，当累加到 FFh 时，下次从 0 开始。传输过程中 Tester 可能会重传 36 消息，ECU 需要根据 ISO 14229 的要求正确处理这种场景的消息传输。为了防止 36 消息无限循环，建议 ECU 内部对 BlockSequenceCounter 重置次数计数，达到一定次数后，返回 NRC 73 的否定响应。

|              |                         |     |                                              |
|--------------|-------------------------|-----|----------------------------------------------|
| Direction 方向 | Tester -> ECU, request  |     |                                              |
| Service 服务   | TransferData            |     |                                              |
| No.序号        | Value 值                 | Cvt | Description 说明                               |
| #1           | 36h                     | M   | Service ID 服务 ID                             |
| #2           | 01h                     | M   | BlockSequenceCounter blockSequenceCounter 序号 |
| #3           | 00-FFh                  | C   | Data 数据                                      |
| ...          | 00-FFh                  | U   | ...                                          |
| #n           | 00-FFh                  | U   | Data 数据                                      |
|              |                         |     |                                              |
| Direction 方向 | ECU -> Tester, Response |     |                                              |
| Service 服务   | TransferData            |     |                                              |
| No.序号        | Value 值                 | Cvt | Description 说明                               |
| #1           | 76h                     | M   | Service ID 服务 ID                             |
| #2           | 01h                     | M   | BlockSequenceCounter blockSequenceCounter 序号 |

#### 4.5.1.4 (a3) Requesting to Exit Downloading Mode/请求退出下载 (\$37)

The Tester sends this message to enable the ECU to exit the current segment programming.  
 Tester 发送此服务使 ECU 退出当前软件段编程。

If there are still segments to be refreshed, the Tester initiates the process of requesting to download files, transmitting data, and requesting to exit downloading mode for the next continuous segment to be programmed.

如果还有软件段要刷新，Tester 将为下一个要编程的连续软件段启动请求下载、传输数据和请求退出下载的流程。

|              |                        |     |                  |
|--------------|------------------------|-----|------------------|
| Direction 方向 | Tester -> ECU, request |     |                  |
| Service 服务   | RequestTransferExit    |     |                  |
| No.序号        | Value 值                | Cvt | Description 说明   |
| #1           | 37h                    | M   | Service ID 服务 ID |

|              |                         |     |                  |
|--------------|-------------------------|-----|------------------|
|              |                         |     |                  |
| Direction 方向 | ECU -> Tester, Response |     |                  |
| Service 服务   | RequestTransferExit     |     |                  |
| No.序号        | Value 值                 | Cvt | Description 说明   |
| #1           | 77h                     | M   | Service ID 服务 ID |

#### 4.5.1.5 (a4) Security Signature Verification/安全签名校验 (\$31 01 DD02)

After all segments in a file are transmitted, this routine control service is used to trigger the ECU to verify the security signature. If the ECU successfully verifies the security signature, a success message is returned, and the ECU returns a positive response message indicating that the verification is successful (the driver file needs to be activated). Otherwise, a positive response message indicating that the security signature verification fails is sent. When the Tester receives a positive response indicating that the verification fails or a negative response message of NRC, the Tester terminates the reprogramming process. 当一个文件中的所有软件段全部传输完毕后，通过例程控制触发 ECU 内部进行安全签名的校验。如果 ECU 验证安全签名成功，则此例程返回成功，ECU 将给出验证成功的肯定响应消息（驱动文件需要激活驱动）。否则，将发送安全签名校验失败的肯定响应消息。当 Tester 收到此消息的校验失败的肯定响应消息或者 NRC 否定响应时，Tester 将终止重编程流程。

For details about the security signature algorithms and process of software packages, see the *Specification of* 《EEA5.0 ECU 软件刷写签名算法规范》.

软件包的安全签名算法和流程，参考《EEA5.0 ECU 软件刷写签名算法规范》。

|              |                         |     |                                                         |
|--------------|-------------------------|-----|---------------------------------------------------------|
| Direction 方向 | Tester -> ECU, request  |     |                                                         |
| Service 服务   | RoutineControl          |     |                                                         |
| No.序号        | Value 值                 | Cvt | Description 说明                                          |
| #1           | 31h                     | M   | Service ID 服务 ID                                        |
| #2           | 01h                     | M   | Start                                                   |
| #3           | DDh                     | M   | 0xDD02 security signature verification<br>0xDD02 安全签名验证 |
| #4           | 02h                     | M   |                                                         |
| #5           | 00-FFh                  | M   |                                                         |
| ...          | 00-FFh                  | M   | Security signature value<br>安全签名的值                      |
| #N           | 00-FFh                  | M   |                                                         |
|              |                         |     |                                                         |
| Direction 方向 | ECU -> Tester, Response |     |                                                         |

| Service 服务 | RoutineControl |     |                                                                                                                                                                                                                                                                                              |
|------------|----------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No.序号      | Value 值        | Cvt | Description 说明                                                                                                                                                                                                                                                                               |
| #1         | 71h            | M   | Service ID 服务 ID                                                                                                                                                                                                                                                                             |
| #2         | 01h            | M   | Start                                                                                                                                                                                                                                                                                        |
| #3         | DDh            | M   | 0xDD02 security signature verification                                                                                                                                                                                                                                                       |
| #4         | 02h            | M   | 0xDD02 安全签名验证                                                                                                                                                                                                                                                                                |
| #5         | 00/01/02/03h   | M   | 00: indicates that the verification is successful 成功<br>01: indicates that the verification of supplier signature fails 供应商签名校验失败<br>02: indicates that the verification of OEM signature fails 主机厂签名校验失败<br>03: indicates that the verification fails because the keys do not exist 密钥不存在失败 |

#### 4.5.1.6 (b) Erasing the Memory/擦除存储器 (\$31 01 FF00)

The Tester sends this routine control message to erase the flash. The flash erasing operation can be executed only after the operation of writing fingerprints successfully into the flash in section 4.3.5. If the file is compressed, MemorySize uses an uncompressed value.

Tesrer 通过发送此例程控制服务进行擦除 Flash 操作，在 4.3.5 写入指纹步骤执行成功并成功把指纹信息写入 Flash 后才允许进行 Flash 的擦除操作。文件如果压缩，MemorySize 使用未压缩的值。

All segments to be programmed in the current file are erased at a time once this operation is executed. After receiving a request for erasing the flash, the ECU sets the application status to invalid and sends a response message.

此步骤将一次性擦除当前文件要编程的所有软件段区域。在收到擦除 Flash 的请求时，ECU 应首先将应用程序状态设置为无效，然后发送响应消息。

| Direction 方向 | Tester -> ECU, request |     |                                                       |
|--------------|------------------------|-----|-------------------------------------------------------|
| Service 服务   | RoutineControl         |     |                                                       |
| No.序号        | Value 值                | Cvt | Description 说明                                        |
| #1           | 31h                    | M   | Service ID 服务 ID                                      |
| #2           | 01h                    | M   | Start 启动                                              |
| #3           | FFh                    | M   | RoutineIdentifier=FF00h: indicates erasing the memory |
| #4           | 00h                    | M   | RoutineIdentifier=FF00h 擦除内存                          |
| #5           | 44h                    | M   | addressAndLengthFormatIdentifier                      |

|              |                         |     |                                                                                                              |
|--------------|-------------------------|-----|--------------------------------------------------------------------------------------------------------------|
|              |                         |     | bit7-bit4 : indicates data length 数据长度<br>bit3-bit0: indicates data start address 数据起始地址                     |
| #6           | 00-FFh                  | M   | memoryAddress Byte#1 (MSB)                                                                                   |
| #7           | 00-FFh                  | M   | memoryAddress Byte#2                                                                                         |
| #8           | 00-FFh                  | M   | memoryAddress Byte#3                                                                                         |
| #9           | 00-FFh                  | M   | memoryAddress Byte#4                                                                                         |
| #10          | 00-FFh                  | M   | memorySize Byte#1 (MSB)                                                                                      |
| #11          | 00-FFh                  | M   | memorySize Byte#2                                                                                            |
| #12          | 00-FFh                  | M   | memorySize Byte#3                                                                                            |
| #13          | 00-FFh                  | M   | memorySize Byte#4                                                                                            |
|              |                         |     |                                                                                                              |
| Direction 方向 | ECU -> Tester, Response |     |                                                                                                              |
| Service 服务   | RoutineControl          |     |                                                                                                              |
| No.序号        | Value 值                 | Cvt | Description 说明                                                                                               |
| #1           | 71h                     | M   | Service ID 服务 ID                                                                                             |
| #2           | 01h                     | M   | Start 启动                                                                                                     |
| #3           | FFh                     | M   | RoutineIdentifier=FF00h: indicates erasing the memory<br>RoutineIdentifier=FF00h 擦除内存                        |
| #4           | 00h                     | M   |                                                                                                              |
| #5           | 00/01h                  | M   | 00: indicates that the flash is successfully erased 成功<br>01: indicates that the flash fails to be erased 失败 |

#### 4.5.1.7 (c) Downloading Application Files/应用文件下载

The downloading process is the same as that of driver files. For details, see sections 4.5.1.2 to 4.5.1.5. 此步骤下载流程跟驱动文件下载流程一致，请参考 4.5.1.2 至 4.5.1.5 步骤。

#### 4.5.2 Process of Downloading Intelligent Component Files/智能部件文件下载流程

The service whose SID is 0x38 is used to transmit and download intelligent component files without the need of downloading driver files. If the intelligent component has driver files, the files are transmitted and downloaded according to section 4.5.1. Figure 9 shows the file downloading process of the FileDownloadStep.

智能部件文件下载，不需要下载驱动文件，直接使用 SID=0x38 进行文件传输下载。如果智能部件有驱动文件则需要参考 4.5.1 章节进行文件的传输下载。FileDownloadStep 的下载流程图见如下 Figure 9 所示。

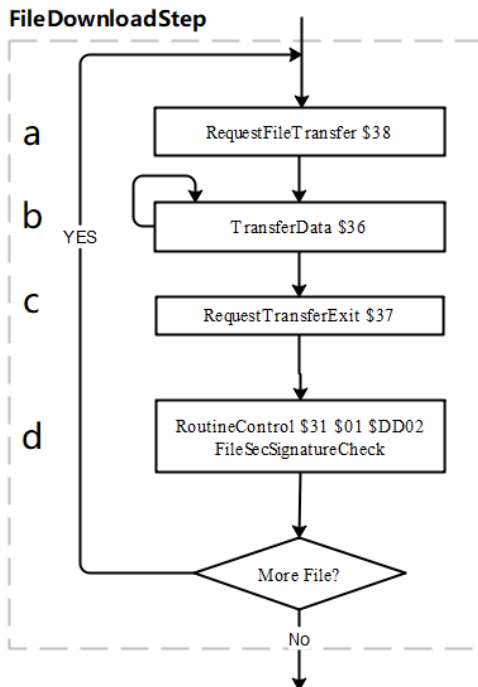


Figure 9: Process of downloading intelligent component files/智能部件文件下载流程

#### 4.5.2.1 (a) Requesting to Transfer File/请求文件传输 (\$38)

For intelligent components that use the service whose SID is 0x38 to transfer files, driver files do not need to be downloaded.

智能部件使用 SID=0x38 进行文件的传输，不需要进行驱动文件下载。

|              |                        |     |                                                                  |
|--------------|------------------------|-----|------------------------------------------------------------------|
| Direction 方向 | Tester -> ECU, request |     |                                                                  |
| Service 服务   | RequestFileTransfer    |     |                                                                  |
| No.序号        | Value 值                | Cvt | Description 说明                                                   |
| #1           | 38h                    | M   | Service ID 服务 ID                                                 |
| #2           | 03h                    | M   | Working mode: replacing files 工作模式：取代文件                          |
| #3           | 00-FFh                 | M   | File path and name length Byte#1 (MSB)<br>文件路径和名称长度 Byte#1 (MSB) |
| #4           | 00-FFh                 | M   | File path and name length Byte#2 (MSB)<br>文件路径和名称长度 Byte #2      |
| #5           | 00-FFh                 | M   | File path and name Byte#1 (MSB)<br>文件路径和名称 Byte#1 (MSB)          |

|                 |                         |     |                                                                                              |
|-----------------|-------------------------|-----|----------------------------------------------------------------------------------------------|
| ...             | 00-FFh                  | C   | ...                                                                                          |
| #5 + n - 1      | 00-FFh                  | C   | File path and name Byte#n<br>文件路径和名称 Byte#n                                                  |
| #5 + n          | 00h                     | M   | Data format:<br>数据格式:<br>00h indicates that the file is not compressed or encrypted 未压缩, 未加密 |
| #5 + n + 1      | 00-FFh                  | M   | Length of the file size parameter 文件大小参数长度                                                   |
| #5 + n + 2      | 00-FFh                  | M   | Uncompressed file size Byte#1 (MSB)<br>文件未压缩大小 Byte#1 (MSB)                                  |
| ...             | 00-FFh                  | C   | ...                                                                                          |
| #5 + n + 1 + k  | 00-FFh                  | C   | Uncompressed file size Byte#k<br>文件未压缩大小 Byte#k                                              |
| #5 + n + 2 + k  | 00-FFh                  | M   | Compressed file size Byte#1 (MSB)<br>文件压缩大小 Byte#1 (MSB)                                     |
| ...             | 00-FFh                  | C   | ...                                                                                          |
| #5 + n + 1 + 2k | 00-FFh                  | C   | Compressed file size Byte#k<br>文件压缩大小 Byte#k                                                 |
|                 |                         |     |                                                                                              |
| Direction 方向    | ECU -> Tester, Response |     |                                                                                              |
| Service 服务      | RequestFileTransfer     |     |                                                                                              |
| No.序号           | Value 值                 | Cvt | Description 说明                                                                               |
| #1              | 78h                     | M   | Service ID 服务 ID                                                                             |
| #2              | 03h                     | M   | Working mode: replacing files 工作模式: 取代文件                                                     |
| #3              | 00-FFh                  | M   | Length format identifier 长度格式标识符                                                             |
| #4              | 00-FFh                  | M   | maxNumberOfBlockLength Byte#1 (MSB)                                                          |
| ...             | 00-FFh                  | C   | ...                                                                                          |
| #4 + m - 1      | 00-FFh                  | C   | maxNumberOfBlockLength Byte#m                                                                |
| #4 + m          | 00h                     | M   | Data format: unencrypted and uncompressed<br>数据格式: 未加密, 未压缩                                  |

#### 4.5.2.2 (b) Transmitting Data/传输数据 (\$36)

For details, see section 4.5.1.3 "Transmitting Data".

详细情况，请参考 4.5.1.3 传输数据。

#### 4.5.2.3 (c) Requesting to Exit Downloading Mode/请求退出下载 (\$37)

For details, see section 4.5.1.4 "Requesting to Exit Downloading".

详细情况，请参考 4.5.1.4 请求退出下载。

#### 4.5.2.4 (d) Security Signature Verification/安全签名校验 (\$31 01 DD02)

For details, see section 4.5.1.5 "Security Signature Verification".

详细情况，请参考 4.5.1.5 安全签名校验。

### 4.6 Requirements for Dual Backup ECUs/双备份控制器的要求

The reprogramming process for ECUs that support dual backup, same as that for common ECUs, consists of pre-programming, programming, and post-programming steps. The operation sequence of pre- and post-programming steps of the dual-backup and common ECUs are the same. The main difference is the operation described in section 4.3.4 and 4.3.7 are optional. For details about the programming step, see section 4.3.

对于支持双备份的 ECU，其重编程规范流程跟一般 ECU 的重编程流程一样，同样分为预编程步骤，编程步骤，后编程步骤。其中预编程步骤和后编程步骤重编程时序是相同的。重编程流程的主要区别在于编程步骤的 4.3.4 步骤和 4.3.7 步骤可选支持，具体流程的详细信息请参考 4.3 编程步骤。

Dual-backup ECUs have two file downloading areas: partitions A and B. The files to be updated in partitions A and B can be different or the same. For dual-backup ECUs that have hardware dual-partitions (also called physical dual-partitions), the upgrade files in partitions A and B can be of the same version, and the ECUs only need to release one software package. For dual-backup ECUs that have software dual-partitions (also called logical dual-partitions), the upgrade files in partitions A and B can be of different versions, and the ECUs need to release software packages A and B.

双备份的 ECU 存在两个文件下载区域，分为“A 区”和“B 区”。A、B 区的刷写文件可以是不同的版本文件，也可以是相同的版本文件。双备份 ECU 属于硬件双分区的（也叫物理双分区），A、B 区的刷写文件可以是相同的版本文件，ECU 只需要发布一个软件包；双备份 ECU 属于软件双分区的（也叫逻辑双分区），A、B 区的刷写文件可以是不同的版本文件，ECU 需要发布 A/B 两个软件包。

Dual-backup ECUs must be able to wait for the master computer to send partition switchover and activation commands (see section 4.3.7 and 4.6.2.5). Non-dual-backup ECUs are automatically activated after being restarted according to the upgrade process.

双备份的 ECU 须支持等待上位机发送切区和激活的指令（见 4.3.7 和 4.6.2.5 章节），无双备份的 ECU 激活按照升级流程重启自激活。

When a dual-backup ECU performs programming in the running partition A or B, the ECU needs to control whether the switchover of the running partition is allowed (see section 4.6.1) and trigger the switchover of the running partitions (see section 4.6.2).

双备份 ECU 刷写 A/B 运行区过程中，需支持控制 ECU 是否允许运行分区倒换（参考 4.6.1 章节）和触发分区倒换（参考 4.6.2 章节）。

Figure 10 shows the partition switchover process when the dual-backup controller does not deliver a command to prevent automatic partition switchover.

双备份控制器刷写过程中未下发阻止自动切区时，运行分区倒换过程应如图 10 所示：

- 1) During the software programming from running partition A to running partition B, the command (\$31 01 DD 0F) for preventing automatic partition switchover is not delivered.  
运行区 A 刷写运行区 B，过程没有下发阻止自动切区命令（\$31 01 DD 0F）；
- 2) The programming is complete and the programming dependency is checked. (For intelligent components, the version installation should be performed.)  
刷写完成并完成编程依赖性检查（对智能部件需完成版本安装）；
- 3) The system is restarted and enters the Bootloader  
重启进入 bootloader；
- 4) The Bootloader guides the system to restart and enter running partition B.  
Bootloader 引导系统启动进入运行分区 B；
- 5) If the operation in step 4 fails, the system rolls back to the Bootloader.  
步骤 4）失败时需回退至 bootloader；
- 6) The Bootloader guides the system to roll back to running partition A.  
Bootloader 引导系统回滚至之前的运行分区 A。

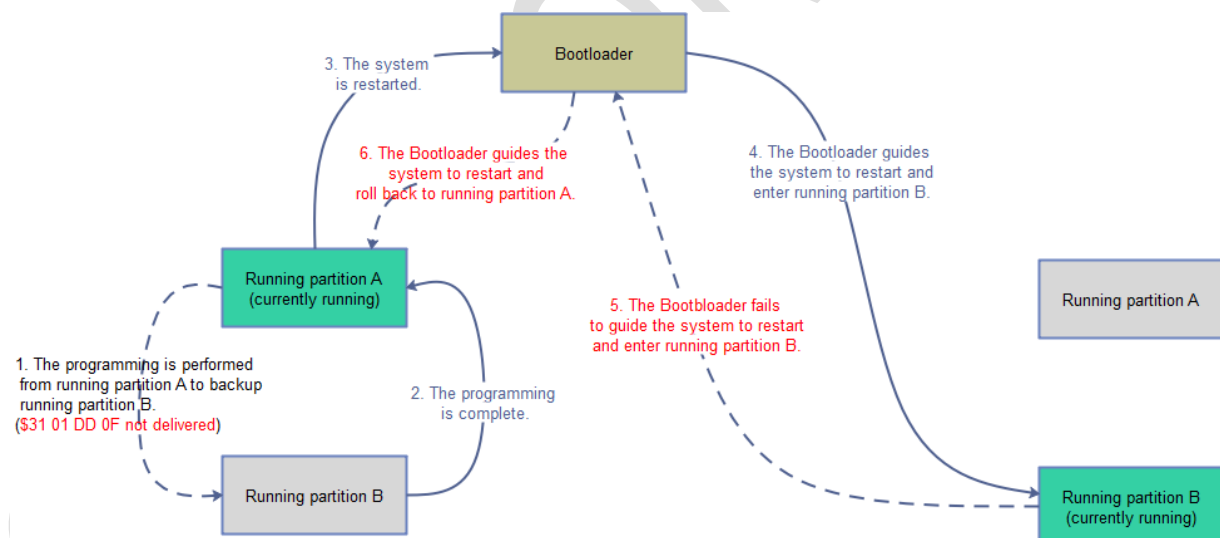


Figure 10: Partition switchover after automatic partition switchover is prevented during partition A/B programming / A/B 区刷写未发“禁止自动切区”后的分区倒换示意图

Figure 11 shows the partition switchover process when the dual-backup controller delivers a command to prevent automatic partition switchover during programming.

双备份控制器刷写过程中下发阻止自动切区时，运行分区倒换过程应如图 11 所示：

- 1) During the software programming from running partition A to running partition B, the command (\$31 01 DD 0F) to prevent automatic partition switchover is delivered after data transmission is complete.

- 运行区 A 刷写运行区 B，数据传输完成后下发阻止自动切区命令（\$31 01 DD 0F）；
- 2) The programming is complete and the programming dependency is checked. (For intelligent components, the version installation should be performed.)  
刷写完成并完成编程依赖性检查（对智能部件需完成版本安装）；
  - 3) The system is restarted and enters the Bootloader.  
重启进入 bootloader；
  - 4) The Bootloader guides the system to restart and enter running partition A.  
Bootloader 引导系统启动进入运行分区 A；
  - 5) Partition switchover process (\$31 01 DD 04 command) is performed. For details, see section 4.6.2.  
切区流程（\$31 01 DD 04 命令，详细流程参考 4.6.2 章节）；
  - 6) The Bootloader guides the system to restart and enter running partition B.  
Bootloader 引导系统启动进入运行分区 B；
  - 7) If the operation in step 6 fails, the system rolls back to the Bootloader.  
步骤 6）失败时需回退至 bootloader；
  - 8) The Bootloader guides the system to roll back to running partition A.  
Bootloader 引导系统回滚至之前的运行分区 A。

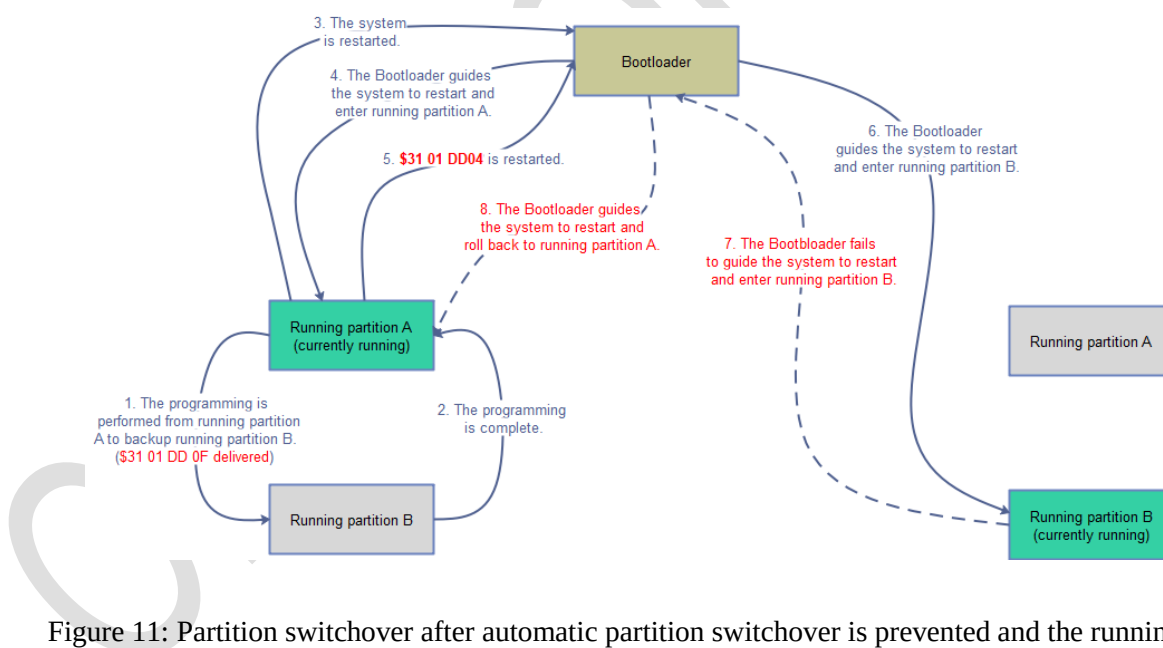


Figure 11: Partition switchover after automatic partition switchover is prevented and the running partition is switched/ A/B 区刷写下发“禁止自动切区”&“切换运行区”后的分区倒换示意图

#### 4.6.1 Partition Switchover Prevention Process of Dual-Backup ECUs/双备份 ECU 运行分区阻止切换流程

To support flexible configuration of vehicle programming policies (such as remote upgrade policies), the dual-backup ECUs need to ensure that the applications in the current running partition are still valid

and reliable after files or programs that are programmed are downloaded. A request for preventing partition switchover needs to be sent to avoid unexpected automatic partition switchover. The running partition switchover of the component is allowed only after the partition switchover command is received again (see section 4.6.2).

为了支持可灵活配置车辆刷写策略（如远程升级策略），双备份 ECU 在刷写文件或刷写程序完成下载后，需保证当前运行分区应用程序仍然有效和可靠，需发送阻止切区请求避免意外的运行分区自动切换动作，仅当再次接收到切区命令后（参考 4.6.2）能够允许部件的运行分区倒换。

Users should define the command for the RoutineControl Data Identifier (0xDD0F) functioning as the client to request the server to prevent automatic partition switchover. For details, see section 4.3.7.

定义 RoutineControl Data Identifier - 0xDD0F 作为客户端请求服务端阻止自动切换运行分区的指令，详细命令可参考 4.3.7 阻止切区步骤。

#### 4.6.2 Partition Switchover Process of Dual-Backup ECUs/双备份 ECU 运行分区切换流程

The dual-backup ECU should support switching from the current program running partition to another partition. The switchover takes effect after the ECU resets. All messages in this process are physical addressing messages. For example, the ECU running in partition A runs in partition B after the running partition switchover and reset, the reverse is also true. If the switchover fails, the ECU should keep running in the partition before the switchover. That is, if the switchover from partition A to partition B fails, the software version in partition A is finally running.

双备份 ECU，应该支持运行分区切换流程，用于将程序运行分区从当前运行分区切换的另外的分区，在复位后生效。此流程所有消息均为物理寻址消息。比如当前是运行在 A 分区，经过此切换流程，ECU 复位后应该运行在 B 分区，反之亦然。如果切换失败，ECU 应该保持在切换前的可运行分区运行，即 A 分区切换到 B 分区切换失败时，最终运行分区是 A 分区软件版本。

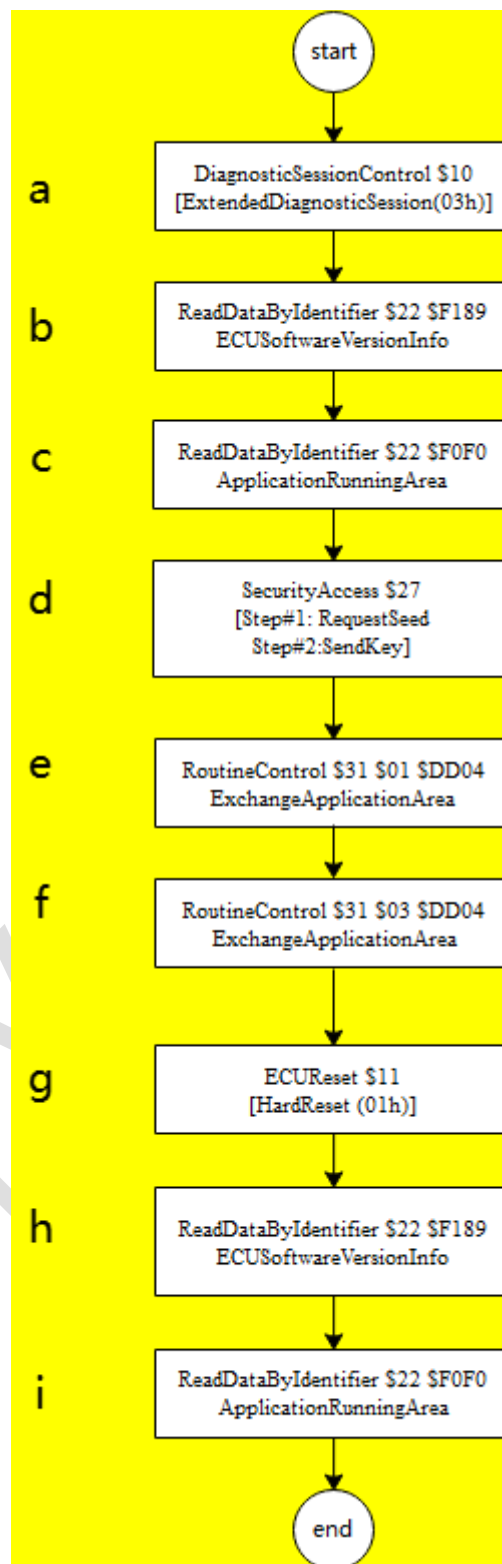


Figure 12: Partition switchover process of dual-backup ECUs/双备份 ECU 运行分区切换流程图

#### 4.6.2.1 (a) Diagnostic Session Control/诊断会话控制(\$10)

The Tester sends a \$10 03 physical addressing message to the ECU and requests the ECU to enter the extended session mode.

发送\$10 03的物理寻址消息，请求进入扩展会话模式。

|              |                          |     |                         |
|--------------|--------------------------|-----|-------------------------|
| Direction 方向 | Tester -> ECU, request   |     |                         |
| Service 服务   | DiagnosticSessionControl |     |                         |
| No.序号        | Value 值                  | Cvt | Description 说明          |
| #1           | 10h                      | M   | Service ID 服务 ID        |
| #2           | 03h                      | M   | Extended session 扩展模式   |
|              |                          |     |                         |
| Direction 方向 | ECU -> Tester, Response  |     |                         |
| Service 服务   | DiagnosticSessionControl |     |                         |
| No.序号        | Value 值                  | Cvt | Description 说明          |
| #1           | 50h                      | M   | Service ID 服务 ID        |
| #2           | 03h                      | M   | Extended session 扩展模式   |
| #3           | 00-FFh                   | M   | P2ServerMax(HightByte)  |
| #4           | 00-FFh                   | M   | P2ServerMax(LowByte)    |
| #5           | 00-FFh                   | M   | P2*ServerMax(HightByte) |
| #6           | 00-FFh                   | M   | P2*ServerMax(LowByte)   |

After the ECU enters the extended session mode, the Tester needs to periodically send a functional addressing message of the TesterPresent (\$3E 80) service to keep the extended diagnostic session active. The sending interval is 2s.

进入扩展会话模式后，Tester 需要发送周期性功能寻址的 TesterPresent (\$3E 80)服务，以保持扩展诊断会话处于活动状态。发送间隔为 2.0 秒。

#### 4.6.2.2 (b) Software Version Information Reading/读取软件版本信息 (\$22 F189)

In this step, the software version information of the partition in which the current application runs is read.

此步骤读取当前程序运行的分区的软件版本信息。

|              |                        |     |                  |
|--------------|------------------------|-----|------------------|
| Direction 方向 | Tester -> ECU, request |     |                  |
| Service 服务   | ReadDataByIdentifier   |     |                  |
| No.序号        | Value 值                | Cvt | Description 说明   |
| #1           | 22h                    | M   | Service ID 服务 ID |

|              |                         |     |                                                                                                                            |
|--------------|-------------------------|-----|----------------------------------------------------------------------------------------------------------------------------|
| #2           | F1h                     | M   | 0xF189 software version                                                                                                    |
| #3           | 89h                     | M   | 0xF189 软件版本号                                                                                                               |
|              |                         |     |                                                                                                                            |
| Direction 方向 | ECU -> Tester, Response |     |                                                                                                                            |
| Service 服务   | ReadDataByIdentifier    |     |                                                                                                                            |
| No.序号        | Value 值                 | Cvt | Description 说明                                                                                                             |
| #1           | 62h                     | M   | Service ID 服务 ID                                                                                                           |
| #2           | F1h                     | M   | 0xF189 software version                                                                                                    |
| #3           | 89h                     | M   | 0xF189 软件版本号                                                                                                               |
| #4           | 00-FFh                  | M   | For details about the format of the 0xF189 software version number, see the system DID definition in the 《奇瑞汽车 ECU 诊断调查表》. |
| ...          | 00-FFh                  | M   |                                                                                                                            |
| #N           | 00-FFh                  | M   | 0xF189 软件版本号格式详细定义, 请参考《奇瑞汽车 ECU 诊断调查表》系统 DID 定义。                                                                          |

#### 4.6.2.3 (c) Current Running Partition Information Reading/读取当前运行分区信息 (\$22 F0F0)

This step is used to read information about the current partition that runs the program.  
 此步骤用于读取当前程序运行的分区信息。

|              |                         |     |                                              |
|--------------|-------------------------|-----|----------------------------------------------|
| Direction 方向 | Tester -> ECU, request  |     |                                              |
| Service 服务   | ReadDataByIdentifier    |     |                                              |
| No.序号        | Value 值                 | Cvt | Description 说明                               |
| #1           | 22h                     | M   | Service ID 服务 ID                             |
| #2           | F0h                     | M   | 0xF0F0 current running partition information |
| #3           | F0h                     | M   | 0xF0F0 当前运行的分区信息                             |
|              |                         |     |                                              |
| Direction 方向 | ECU -> Tester, Response |     |                                              |
| Service 服务   | ReadDataByIdentifier    |     |                                              |
| No.序号        | Value 值                 | Cvt | Description 说明                               |
| #1           | 62h                     | M   | Service ID 服务 ID                             |
| #2           | F0h                     | M   | 0xF0F0 current running partition information |
| #3           | F0h                     | M   | 0xF0F0 当前运行的分区信息                             |

|    |        |   |       |
|----|--------|---|-------|
| #4 | 41/42h | M | “A/B” |
|----|--------|---|-------|

#### 4.6.2.4 (d) Security Access/安全访问 (\$27)

Secure authentication is used for routine control. The \$27 01/02 service is used for security authentication. For details about the security authentication process and algorithms, see the *Design Specification of* 《EEA5.0 ECU 27 服务算法规范》.

例程控制需要进行安全访问解锁。此使用\$27 01/02 服务用于安全解锁。此安全解锁具体流程和算法请参考《EEA5.0 ECU 27 服务算法规范》文档。

#### 4.6.2.5 (e) Version Switchover/版本切换 (\$31 01 DD04)

After receiving this routine control message, the ECU should switch to the currently activated partition and make it take effect after the reset. The version switchover process takes a maximum of 30 minutes. ECU 在收到此例程消息后，应该切换当前激活的分区，并在复位后生效。版本切换流程耗时不能超过 30 分钟。

|              |                         |     |                                                                                                                                                            |
|--------------|-------------------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direction 方向 | Tester -> ECU, request  |     |                                                                                                                                                            |
| Service 服务   | RoutineControl          |     |                                                                                                                                                            |
| No.序号        | Value 值                 | Cvt | Description 说明                                                                                                                                             |
| #1           | 31h                     | M   | Service ID 服务 ID                                                                                                                                           |
| #2           | 01h                     | M   | Start 启动                                                                                                                                                   |
| #3           | DDh                     | M   | 0xDD04 version switchover                                                                                                                                  |
| #4           | 04h                     | M   | 0xDD04 版本切换                                                                                                                                                |
|              |                         |     |                                                                                                                                                            |
| Direction 方向 | ECU -> Tester, Response |     |                                                                                                                                                            |
| Service 服务   | RoutineControl          |     |                                                                                                                                                            |
| No.序号        | Value 值                 | Cvt | Description 说明                                                                                                                                             |
| #1           | 71h                     | M   | Service ID 服务 ID                                                                                                                                           |
| #2           | 01h                     | M   | Start 启动                                                                                                                                                   |
| #3           | DDh                     | M   | 0xDD04 version switchover                                                                                                                                  |
| #4           | 04h                     | M   | 0xDD04 版本切换                                                                                                                                                |
| #5           | 00-02h                  | M   | 00: indicates that the switchover is successful 成功<br>01: indicates that the switchover fails 失败<br>02-FFh: indicates other error codes, which are defined |

|  |  |  |                                                                                         |
|--|--|--|-----------------------------------------------------------------------------------------|
|  |  |  | by the supplier and described in the diagnostic question-<br>naire 其他错误码，供应商定义并在诊断调查表反馈 |
|--|--|--|-----------------------------------------------------------------------------------------|

#### 4.6.2.6 (f) Version Switchover Results Querying/查询版本切换结果 (\$31 03 DD04)

During version switchover, the ECU must respond according to the actual progress of the current version switchover after receiving this routine message. After sending the request for querying the version switchover results, the Tester invokes this service every 1s to query the version switchover results. If the ECU does not respond to this message within the timeout period, the Tester cannot process the message as an error. Instead, the Tester needs to continue querying the results until the final results status is obtained. The ECU may fail to complete the version switchover due to a fault. Therefore, a timeout period should be set in this step. If the timeout period T2 is 30 minutes and the query continues until the timeout, the Tester stops the version switchover process.

ECU 在版本切换过程中，在接收到此例程消息后，须按照当前版本切换实际进展进行响应。

Tester 在发送 (f) 步骤后，须以 1s 周期调用此服务，查询版本切换结果。ECU 超时未响应此消息，Tester 不能当做错误处理，应该继续查询，直到获取到最终的结果状态。ECU 可能由于故障原因一直无法完成版本切换，此步骤应该设置超时时间，超时时间 T2=30 分钟没有完成或者结束，Tester 应该终止本次版本切换流程。

|              |                         |     |                                                                 |
|--------------|-------------------------|-----|-----------------------------------------------------------------|
| Direction 方向 | Tester -> ECU, request  |     |                                                                 |
| Service 服务   | RoutineControl          |     |                                                                 |
| No.序号        | Value 值                 | Cvt | Description 说明                                                  |
| #1           | 31h                     | M   | Service ID 服务 ID                                                |
| #2           | 03h                     | M   | Request Routine Results 请求例程结果                                  |
| #3           | DDh                     | M   | 0xDD04 version switchover                                       |
| #4           | 04h                     | M   | 0xDD04 版本切换                                                     |
|              |                         |     |                                                                 |
| Direction 方向 | ECU -> Tester, Response |     |                                                                 |
| Service 服务   | RoutineControl          |     |                                                                 |
| No.序号        | Value 值                 | Cvt | Description 说明                                                  |
| #1           | 71h                     | M   | Service ID 服务 ID                                                |
| #2           | 03h                     | M   | Routine results query 请求例程结果                                    |
| #3           | DDh                     | M   | 0xDD04 version switchover                                       |
| #4           | 04h                     | M   | 0xDD04 版本切换                                                     |
| #5           | 00-02h                  | M   | 00: indicates that the version switchover is successful<br>切区完成 |

|  |  |  |                                                      |
|--|--|--|------------------------------------------------------|
|  |  |  | 01: indicates that the version switchover fails 切区失败 |
|  |  |  | 02: indicates that the version is being switched 切区中 |

#### 4.6.2.7 (g) ECU Reset 复位 (\$11)

This message is used to restart the ECU. After receiving the message, the ECU should actively disable the socket communication of the current diagnostic link (only for the DoIP link) before the reset. After sending this message, the Tester restores the link to the ECU (for details, see section 4.3.2), and then performs subsequent operations.

此服务将使 ECU 重启。ECU 在收到此消息复位前，应该主动的关闭当前诊断链路的 Socket 通讯（仅仅对 DoIP 的链路有此要求）。Tester 在发送完此命令后，参考 4.3.2 步骤恢复和 ECU 的链路后，再执行后续步骤。

|              |                         |     |                  |
|--------------|-------------------------|-----|------------------|
| Direction 方向 | Tester -> ECU, request  |     |                  |
| Service 服务   | ECUReset                |     |                  |
| No.序号        | Value 值                 | Cvt | Description 说明   |
| #1           | 11h                     | M   | Service ID 服务 ID |
| #2           | 01h                     | M   | Hard reset 硬复位   |
|              |                         |     |                  |
| Direction 方向 | ECU -> Tester, Response |     |                  |
| Service 服务   | ECUReset                |     |                  |
| No.序号        | Value 值                 | Cvt | Description 说明   |
| #1           | 51h                     | M   | Service ID 服务 ID |
| #2           | 01h                     | M   | Hard reset 硬复位   |

#### 4.6.2.8 (h) (i) Post-Reset Information Reading/读取复位后的信息

The two steps are mainly used to read the version information of the program software after the ECU reset and the information about the currently running partition to determine whether the version switchover is successful. If the program partition information and software version information read after the ECU reset are the same as those before the ECU reset, the switchover fails. For details, see sections 4.6.2.2 and 4.6.2.3.

此两个步骤主要用于程序复位后，读取复位后的程序软件版本信息和当前运行的分区信息用于判断此次版本切换流程的最终结果是否成功。如果在复位后读取到的程序分区信息以及软件版本信息和复位前一样，则切换失败。具体信息请参考 4.6.2.2 步骤和 4.6.2.3 步骤。

#### 4.6.3 Reference Solution for Preventing and Restoring Automatic Partition Switchover of Dual-Backup ECUs/双备份 ECU 禁止自动切区和恢复自动切区实现参考方案

This section describes the reference implementation solution of the ECU, to help developers understand the requirements of dual-backup programming specifications for intelligent vehicles. The specific implementation solution of the ECU is not mandatory in this specification. Figure 13, Figure 14 and Figure 15 are the time sequence diagram that show a reference solution for updating and restarting the partition enablement flag bit and partition switchover disablement status bit of a dual-backup ECU.

本章节为控制器的参考实现方案，目的是方便开发者理解智能汽车的双备份刷写规范的需求，控制器的具体实现方案本规范不做强制要求。图 13，图 14 和图 15 是双备份 ECU 的启动分区标识位和禁止切区状态位的更新机制和重启的参考方案时序图。

This reference solution is designed based on the following prerequisites:  
本参考方案的三个设计前提说明：

- 1) In this document, the automatic partition switchover action is defined as that the ECU automatically updates the partition enablement flag bit after a certain condition is met. If the programming dependency check is successful, the ECU considers that the programming is successful and updates the partition enablement flag bit. In this case, the automatic partition switchover action is included in the programming dependency check step.

本规范中所述自动切区动作，定义为 ECU 在某个条件满足后自动更新启动分区标志位。如编程依赖性检查成功后，ECU 即可认为刷写成功即可更新启动分区标志位，此时自动切区动作包含在编程依赖性检查步骤中。

- 2) The ECU shall maintain a Boolean value **FrbExchgAreaFlag**, which disables automatic partition switchover. If **FrbExchgAreaFlag** is set to False, the ECU does not disable automatic partition switchover. If **FrbExchgAreaFlag** is set to True, the ECU disables automatic partition switchover. The default value of the status bit **FrbExchgAreaFlag** is False and loss upon power-off is allowed.

ECU 应维护一个禁止自动切区的状态 Boolean 值 FrbExchgAreaFlag。当 FrbExchgAreaFlag=False 时，表示 ECU 未开启阻止自动切区；当 FrbExchgAreaFlag=True 时，表示 ECU 开启阻止自动切区。状态位 FrbExchgAreaFlag 的默认值为 False 且允许掉电丢失。

- 3) The ECU should maintain a partition enablement flag bit **StrtRngAreaFlag**. If **StrtRngAreaFlag** is set to A, the ECU needs to run in partition A when the ECU is restarted next time. If **StrtRngAreaFlag** is set to B, the ECU needs to run in partition B when the ECU is restarted next time. The partition enablement flag bit must be stored in the non-volatile storage, and its value cannot be lost upon power-off.

ECU 应维护一个启动分区标识位 StrtRngAreaFlag。当 StrtRngAreaFlag=A，表示下次重启 ECU 需要在 A 分区运行；当 StrtRngAreaFlag = B，表示下次重启 ECU 需要在 B 分区运行。启动分区标识位应存储在非易失性存储区，其值不允许掉电丢失。

The implementation of this solution is described as follows:  
该方案下的三个场景实现说明：

|                                                 |                  |                  |
|-------------------------------------------------|------------------|------------------|
| Revision: 1.3                                   | Department: IVTC | Date: 2022-10-12 |
| Status: Issue                                   |                  | Page 61 of 65    |
| Filename: EEA5.0 ECU bootloader 及刷新规范_V1.3.docx |                  |                  |

- 1) Scenario 1: The command for preventing automatic partition switchover is received. After the ECU receives the command for preventing partition switchover and successfully responds to the command, **FrbExchgAreaFlag** must be set to **True**. Otherwise, **FrbExchgAreaFlag** remains **False**.

场景一：接收禁止自动切区命令场景。当 ECU 接收到禁止切区指令并能够成功响应后，FrbExchgAreaFlag 须置为 True，否则保持为 False。

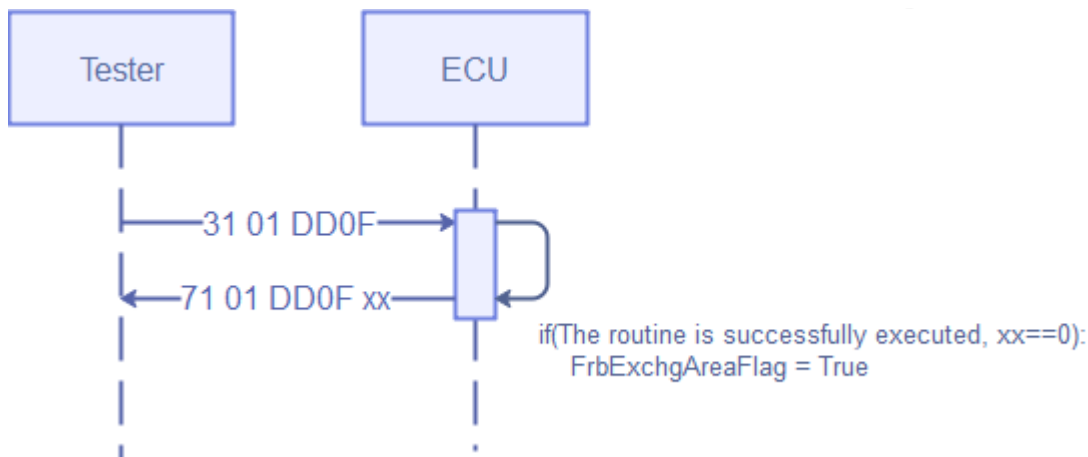


Figure 13: Scenario of assigning values to the status values of the dual-backup ECU preventing partition switchover/双备份 ECU 禁止切区状态值赋值场景

- 2) Scenario 2: A programming dependency check command is received. In this scenario, the automatic partition switchover action is included in the programming dependency check step, as described in prerequisite 1. When the ECU receives the programming dependency check and responds successfully, the ECU determines whether to automatically update the start flag **StrtRngAreaFlag** based on the value of **FrbExchgAreaFlag**. When **FrbExchgAreaFlag** is set to True, the current value of **StrtRngAreaFlag** must be retained and **FrbExchgAreaFlag** must be set to False. When the value of **FrbExchgAreaFlag** is False, **StrtRngAreaFlag** must be set to the programming partition.

场景二：接收编程依赖性检查命令场景，该场景下，如设计前提 1) 所述自动切区动作包含在编程依赖性检查步骤中。当 ECU 接收到编程依赖性检查并能够成功响应，须依赖禁止切区状态值 FrbExchgAreaFlag 判定是否需要执行自动更新启动标识位 StrtRngAreaFlag 的动作。当 FrbExchgAreaFlag=True 时，须保持 StrtRngAreaFlag 当前的值，并重置 FrbExchgAreaFlag=False；当 FrbExchgAreaFlag=False 时，须将 StrtRngAreaFlag 值置为刷新区。

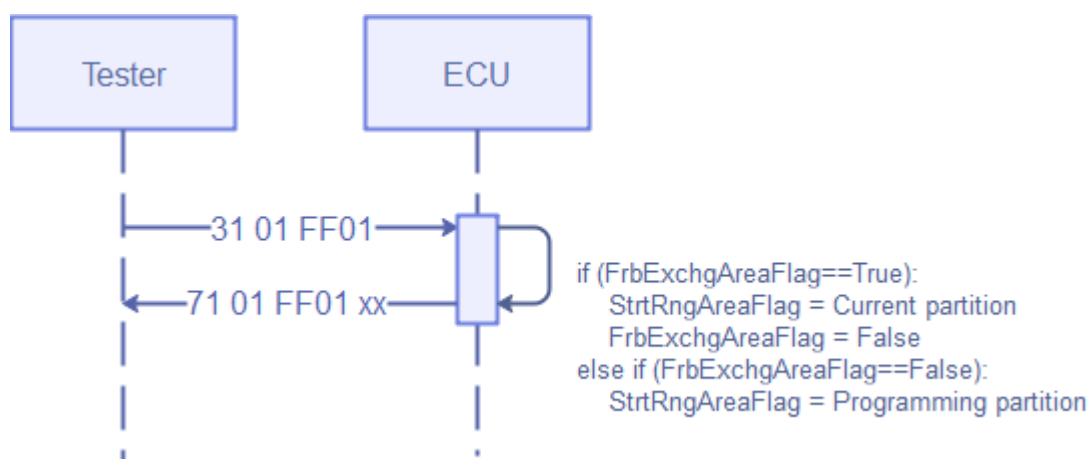


Figure 14: Scenario of automatic partition switchover prevention of dual-backup ECUs (the automatic partition switchover action is performed in the programming dependency check step)/双备份 ECU 禁止自动切区场景-自动切区动作在编程依赖性检查步骤中

- 3) Scenario 3: The partition switchover command is received. In this scenario, forcible partition switchover does not depend on the value of **FrbExchgAreaFlag**. Users can forcibly switch the partition enablement flag to the peer partition. If the ECU can respond successfully, the partition enablement flag **StrtRngAreaFlag** needs to be updated to the corresponding partition. Otherwise, the flag bit must remain as the current running partition.

场景三：接收切区命令场景。该场景下强制切区不依赖于禁止切区状态值 **FrbExchgAreaFlag** 的判定，可以强制切换启动分区标识位为对区。如果 ECU 能够成功响应，则需将启动分区标识 **StrtRngAreaFlag** 更新至对分区，否则该标识位须保持为当前运行分区。

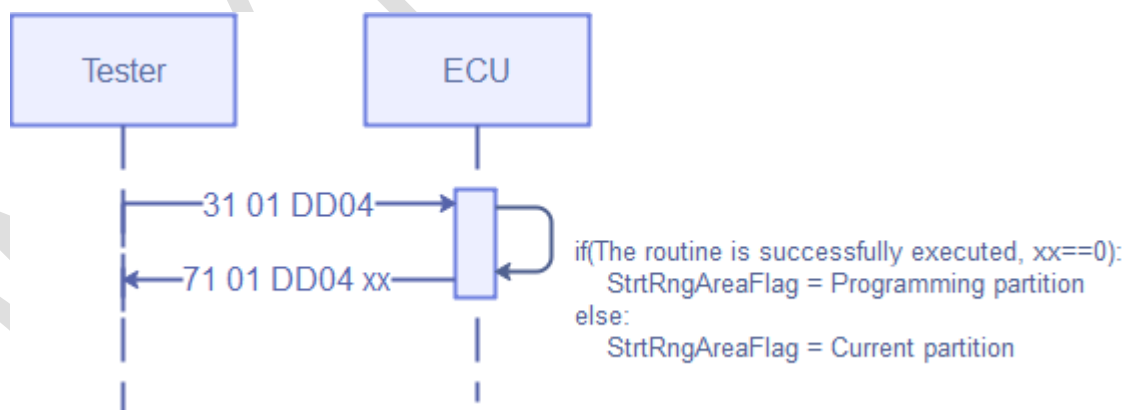


Figure 15: Scenario of partition switchover command received by dual-backup ECUs/双备份 ECU 接收切区指令场景

## Appendix A: RIDs and DID/附录 A: 例程 ID 和 DID

### A.1 DID Information/DID 信息

| DID  | Name 名称                                                   | Format Definition 格式定义                                                                     |
|------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------|
| F184 | Software fingerprint 软件指纹                                 | For details, see section 4.3.5<br>参考: 4.3.5 章节                                             |
| F189 | Software version 软件版本                                     | For details, see the definition of DID in the 《EEA5.0 ECU 诊断调查表》<br>参考: 《EEA5.0 ECU 诊断调查表》 |
| F0F0 | Current running partition information 当前运行分区信息            | For details, see section 4.3.4<br>参考: 4.3.4 章节                                             |
| F0F1 | Number of programming attempts 尝试编程次数计数                   | For details, see section 4.3.5<br>参考: 4.3.5 章节                                             |
| F0F2 | Obtaining the installation progress and results 获取安装进度和结果 | For details, see section 4.3.10<br>参考: 4.3.10 章节                                           |
| F0F3 | Number of programming successes 编程依赖检查成功次数计数              | For details, see section 4.3.8<br>参考: 4.3.8 章节                                             |

### A.2 RID Information/RID 信息

| RID  | Name 名称                                | Type 类型  | Involved Section 涉及章节                                                                                                                                                                             |
|------|----------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0203 | Programming condition check 编程条件检查     | Start 启动 | For details, see section 4.2.3<br>参考: 4.2.3 章节                                                                                                                                                    |
| FF00 | Erasing the memory 擦除存储器               | Start 启动 | For details, see section 4.5.1.6<br>参考: 4.5.1.6 章节                                                                                                                                                |
| FF01 | Programming dependency check 编程依赖关系检查  | Start 启动 | For details, see section 4.3.8<br>参考: 4.3.8 章节                                                                                                                                                    |
| DD01 | StayInBoot                             | Start 启动 | For details, see section 2.2.1<br>参考: 2.2.1 章节<br><br>Note: The application does not support this RID.<br><br>The Bootloader must support this RID.<br>注: APP 不支持此例程 ID<br>BootLoader 内必须支持此例程 ID |
| DD02 | Security signature verification 安全签名校验 | Start 启动 | For details, see section 4.5.1.5<br>参考: 4.5.1.5 章节                                                                                                                                                |
| DD03 | Version installation 版本安装              | Start 启动 | For details, see section 4.3.9<br>参考: 4.3.9 章节                                                                                                                                                    |
| DD04 | Version switchover                     | Start 启动 | For details, see section 4.6.2.5                                                                                                                                                                  |

|      |                               |          |                                                                 |
|------|-------------------------------|----------|-----------------------------------------------------------------|
|      | 版本切换                          |          | 参考：4.6.2.5 章节                                                   |
| DD0F | Switchover prevention<br>阻止切区 | Start 启动 | For details, see section 4.3.7 and 4.6.1<br>参考：4.3.7 和 4.6.1 章节 |